Psalms Asante-Samuels

Interactive Sound and Visitor Engagement:

Exploring Audio interaction and
Immersive Sound in Exhibition Spaces

Audio Interaction within the Being Human Exhibition

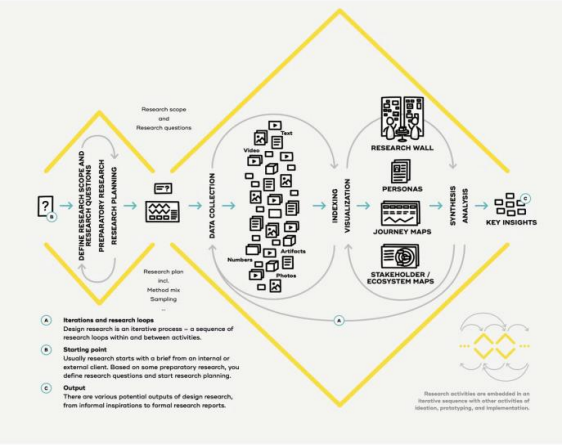
This project is based on research conducted at the **Welcome Colection**. More specifically, in the **Being Human exhibition**; exploring themes around the **human body, health, illnesses and identity** through a range of **interactive** and **immersive** displays.

How? you may ask.

Well, this exhibition combines **physical artefacts, digital installations and sensory experiences** to **engage visitors** in understanding what it means to be human. Therefore, making a strong case study for investigating **how interaction, particularly through sound and immersive elements can influence visitor engagement and experience within their exhibition space.**

This project also presents the **required techniques for a life-centred User Experience design approach**. By drawing on theories and practices from design research, I will examine **cultures, behaviours, systems and artefacts that shape human experience across physical, digital and hybrid environments.**

Focusing on **the first diamond of the Double Diamond framework: Discover and Define**, I will be emphasising the importance of research and definition through **observing, analysing and synthesising experience** rather than redesigning anything.



Screenshot taken from a slide presentation created by my tutors which was taken from **Stickdorn, et al. (2018).**

A larger image will be on the **appendix page**.

Initial Curiosity and Topic Development

After going through the project brief in **week 1-2**, I visited the **Being Human Exhibit** to get a **personal ethnography** for what the museum is like. Making **observations** leading to my topic to be focused **initially on the interactivity of exhibitions and how the relation of the theme identity is used to enhance visitor engagement**.

Providing me with these **5 search terms** I used for my secondary research:

- **Interactive museum engagement identity**
- **Identity representation in museums**
- **Embodied interaction museum experience**
- **Visitor mean-making interactive exhibitions**
- **Participatory exhibition design and identity**

In **week 3**, I posted on the **class Padlet showing the relevance to the readings provided to me by my tutors linked to research**. I stated; linking to **Boylston, S. Designing with society**; design should **create conditions for people to be and to do** and to **connect multiple perspectives together** like what the exhibition does with its **interactive displays and artefacts** which I felt linked more to my focus during that period.

Additionally, I referenced **Booth, et al. The Craft of Research (2016)** and how it teaches that **broad topics should be narrowed into a specific focused issue to strengthen clarity and depth.** However, I never applied the knowledge given at the time.

With the 5 search terms, I found **5 sources for my secondary research** and then produced **5 different research questions using Booth's Framework.**

Briefly about:

- **How interactive exhibition design influence visitor engagement.**
- **How exhibitions use identity-related narratives.**
- **How sound, visuals and spatial immersion influence visitor engagement.**
- **How sound environments and visitor-controlled audio systems influence movement, attention and engagement.**
- **How interactive displays can both engage visitors and potentially trigger emotional discomfort.**

Through **feedback from my tutor**, I realised these were **too broad**....

Booth, W. and Stickdorn, et. al.

The Craft of Research by Booth, W. (2016)

In Chapter 3, Booth explains how researchers transform broad interests into clear, focused research questions. Emphasising that a topic on its own is too general and must be **refined into a specific issue** presenting a problem or gap in understanding.

From there, the researched concluded that **research questions** need to be formulated to identify what is not yet fully understood and framing it in a way to invite investigation.

A strong research question gives direction, purpose and coherence to one's research.

- Strong research questions explain **how and why**.
- Research refines **thinking through writing**.
- Research responds to **meaningful questions**.
- Topics must **narrow into specific issues**.

This is Service Design Doing, Stickdorn, et. al. (2018)

In Chapter 5, practical approaches to conducting within service design are focused on. It highlights the importance of engaging with real contexts, stakeholders and systems to help understand how services function in everyday life. Emphasising, **collaborative research methods, observation and participations as ways to uncover lived experiences.**

Research is presented as **an iterative and reflective** process that deepens understanding before shaping interventions. Therefore, **encouraging primary research to deepen critical knowledge on research.**

- Service design research centres on **real world contexts**.
- Stakeholder engagement is essential for **insight.**
- Observation and participation **reveal lived experiences**.
- Collaboration supports **deeper understanding**.
- Research is iterative and evolves through **reflection**.

Designing with Society, Boylston, S. (2019)

Introduction (pp. 14–23)

In the introduction, Boylston explains that design should address complex social and systemic challenges rather than isolated issues. She introduces the capabilities approach, which focuses on improving what people **are able to do and experience within society.**

Design is framed as a social and ethical practice that influences relationships, participation, and access to opportunities.

Boylston argues that designers are **responsible for shaping conditions that impact human wellbeing and everyday life**.

- Design responds to **social and systemic** challenges
- The capabilities approach focuses on **human opportunity and wellbeing**
- Design shapes **systems, relationships, and participation**
- Designers hold **social and ethical responsibility**
- Design influences everyday **human experience**

Section 1 (pp. 24–27)

In this section, Boylston connects systems thinking with social innovation, explaining that contemporary **issues are interconnected across wider networks and structures**.

She argues that designers must **move beyond isolated problem-solving and instead understand relationships between systems, people, and environments.**

The section highlights how **design decisions can create wider consequences**, reinforcing the need for contextual and long-term thinking when addressing complex challenges.

- Social problems exist **within interconnected systems**
- Systems thinking **focuses on relationships and wider impacts**
- Linear problem-solving **overlooks complexity**
- Design decisions create **long-term consequences**
- Innovation requires **contextual awareness**

Designing with Society, Boylston, S. (2019)

Holistic definition in context:

To look at the **whole system** rather than just one **small part**

Chapter 1: Understanding the Interconnected Nature of our Challenge (pp. 28–38)

In this chapter**, Boylston explores how economic, environmental, and political systems are deeply interconnected, influencing one another across global and local contexts.**

She explains that designers must recognise how actions within one area can create **ripple effects** throughout wider systems.

The chapter encourages a **holistic approach to design**, positioning it as part of a **dynamic network** of relationships rather than an isolated activity. This perspective supports more **responsible and ethical decision-making.**

- Global challenges operate across **interconnected systems**
- Actions create **ripple effects throughout wider networks**
- Designers must **adopt holistic perspectives**
- Design exists within **dynamic relationships and systems**
- Systems awareness supports **ethical decision-making**

Chapter 6: Rejecting the Short for the Long

In this chapter, Boylston argues that society often prioritises short-term thinking over long-term resilience and sustainability.

She explains that deeply **rooted beliefs can make people resistant to change**, limiting their ability to adapt to uncertainty. Drawing on ideas such as **antifragility, pace layer thinking, and ecological systems**, Boylston emphasises that resilient systems are those capable of evolving through disruption rather than resisting it.

The chapter ultimately positions **design as a tool for supporting long-term systemic transformation and sustainable futures.**

- Short-term thinking **weakens long-term resilience**
- Deeply held **beliefs often resist change**
- Antifragile systems **improve through uncertainty/disruption**
- Pace layer thinking explains **systems operating at different speeds**
- Design can **support long-term systemic transformation**

Secondary Research

International Journal of Heritage Studies

By applying knowledge from the **Craft of Research by Booth, et al. – narrowing topics to specific questions**; the importance of this will be displayed through **citations, key ideas, quotes** and **connection to my research** then a **reflection paragraph**.

(Some will be presented on different slides but referenced back to here).

Digital storytelling for emotional engagement in museums: *design and evaluation of the Hunterian Antonine Wall EMOTIVE experiences*. International Journal of Heritage Studies, pp.1–32.

Economou, M., Young, H. and Sosnowska, E. (2025).

- Relatable narratives are built around universal themes (love, death, family etc.)
- Empathetic approaches create deeper resonance with audience

Authority:
This article is published in the **International Journal of Heritage Studies**, which is a **peer reviewed journal** that focuses on museum and heritage research. The authors are researchers that work in digital heritage and museum storytelling. This strengthens the credibility of the findings.

O'Leary (2021) explains in **The Essential Guide to Doing Your Research Project** that "secondary research allows researchers to understand what is already known, identify gaps in knowledge, and position their own enquiry within wider debates" (pg. 89). Without this foundation, primary research risks being unfocused, repetitive, or disconnected from existing understanding.

Emerging Question:
Do museums fully consider how emotionally sensitive identity themes **(e.g., HIV, illness, trauma)** may impact various identity groups differently?

Quote:
"We assessed objects' historical/archaeological significance and ability to connect with universal themes such as death, love and loss."
(Perry, Katifori et al., 2017, 6.2)

Insight
This shows that emotive object selection is intentional — engagement is designed, not accidental.

The Diary of Niels

The Diary of Niels: *Affective engagement through tangible interaction with museum artifacts*. arXiv.

Schou, M. and Løvlie, A. (n.d.).

- Tangible interaction increases engagement
- Hands-on artefacts improve memory recall
- Participatory exhibits increase visitor satisfaction

Authority:
This research paper examines tangible interaction in museum contexts. The authors specialise in interactive systems and cultural technology.

Emerging Question:
How necessary is physical interactivity for an exhibition's success? Can identity engagement occur without visitor-controlled interaction?

Quote:
"In interviews with the guests they were all able to recall and explain the function of at least one of the three historical objects..."

Insight
This demonstrates measurable engagement through recall and interpretation.

International Archives of the Photogrammetry

Embodied Reconstruction of Digital Museum Applications. The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences.

Wang, F. and Du, J. (2025).

- Immersive technology increases emotional resonance
- Multi-sensory environments deepen engagement
- Sound and visuals enhance story-telling

Authority:
This paper is published in the **International Archives of the Photogrammetry**, Remote Sensing and Spatial Information Sciences, an **academic publication focusing on digital environments and spatial technologies.**

Emerging Question:
Does sensory immersion affect all identity groups equally, or might embodied digital interaction amplify engagement differently depending on lived experience?

Quote:
"A high-quality museum experience is vital for sparking public awareness and emotional resonance with cultural heritage."

Insight
The emphasis is on sensory augmentation to deepen emotional and cognitive engagement.

Materialising Contexts

Materialising contexts: *virtual soundscapes for real-world exploration*. Personal and Ubiquitous Computing.

Cliffe, L., Mansell, J., Greenhalgh, C. and Hazzard, A. (2020).

- Audio environments encourage exploration
- Sound can personalise museum experience
- Audio prompts guide visitor movement

Authority:
This research is published a **peer reviewed journal called Personal and Ubiquitous Computing;** this specialises in interactive technologies and human-computer interaction.

Emerging Question:
Does sound augmentation increase engagement across all identity groups? How does accessibility **(e.g., hearing impairments)** complicate identity-based audio immersion?

Quote:
"...suggests other nearby artworks to the visitor that may be of interest to them via spatially located audio prompts."

Insight
This introduces identity-based personalisation into engagement design.

Participatory Exhibition Design

Radical Change: *Participatory Exhibition Design*. Museum-iD.

FIWebteam (2017).

- Visitor-centric exhibitions increase satisfaction
- Participation strengthens connection to artefacts
- Interactive systems encourage exploration

Authority:
This article is **written by museum professional and published by Museum-iD**, they are an organisation that specialise in museum innovation and exhibition design. It **may not be a peer-reviewed academic research source, but it provides very valuable insight into professional museum practice.**

Emerging Question:
How do these layered interactions function as a system? How does participatory design foster mean-making across diverse identity positions?

Quote:
"Participatory projects offer ways to engage members of our community in more innovative ways and can contribute to change the community's perception of the institution as a more inclusive environment."

Insight
Participation here is framed not only as interaction, but as institutional identity transformation.

Reflection For and Against

This research cycle **clarified that participation, and interactivity significantly increases engagement and mean-making within exhibitions**. I learned that **embodied interaction could take various forms - sound, visuals, tactile interfaces** - they each prioritise different **engagement metrics and sensory experiences**.

Additionally, it became evident that museums often draw **on visitors' sense of identity to strengthen connection,** whether through **inclusion, memory, nostalgia or personal experience**, though this may also **introduce emotional sensitivity for some groups**.

Lastly, I recognised that **engagement operates on multiple levels**; through **extended dwelling times, social conversation or even observing others interacting**. All these modes suggest that interaction in the exhibitions **functions as a layered system rather than a single measurable behaviour**.

However, there are **key contradictions and gaps across my sources**. Most of the literature treats **interactivity as inherently positive**, rarely questioning who it may exclude or overwhelm — particularly when dealing **with emotionally sensitive themes**. For example, **the jukebox in the Being Human exhibition** plays catchy, engaging music, yet the songs reference deadly diseases and pandemics.

While this may appear entertaining, it **could trigger discomfort for visitors** who have lost someone to those illnesses. Similarly, the HIV blood artefacts may be perceived as fascinating and powerful, but they may also **resonate deeply or painfully for certain identity groups**.

Although **identity is used strategically within exhibitions to enhance engagement**, there is **limited critical examination** of how different **identity positions shape interpretation and emotional response**. Immersive sensory design is frequently framed as inclusive innovation; however**, accessibility and psychological safety are rarely foregrounded**. This reveals a gap in understanding how interactive systems may shape identity engagement unevenly across diverse visitors.

Pattern Recognition with my Research

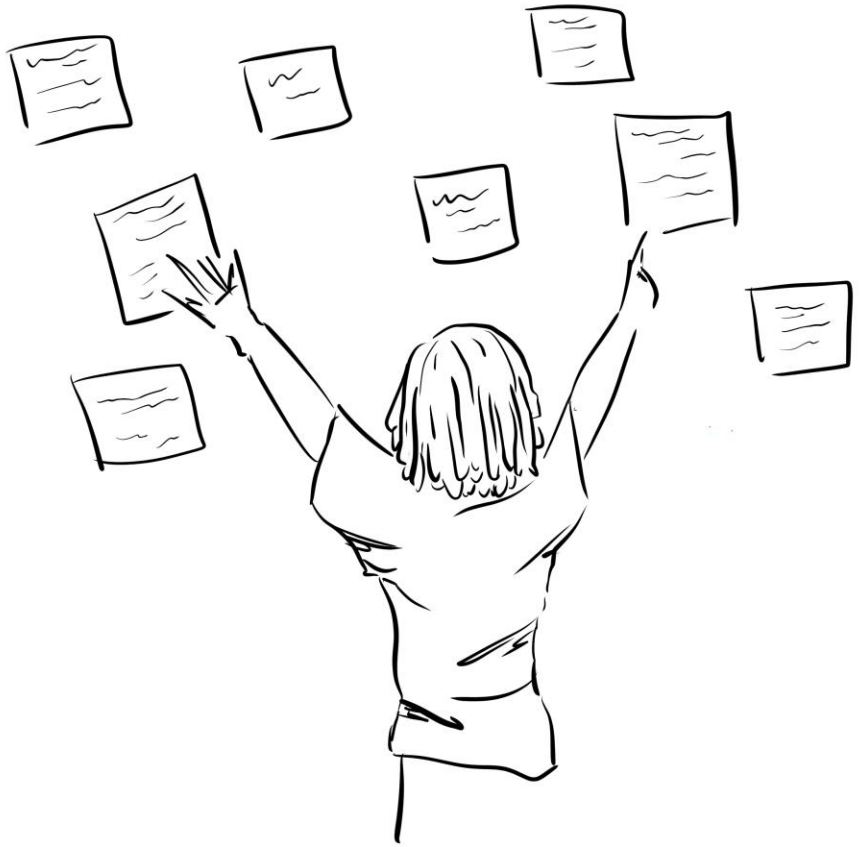
There are various recurring patterns across the sources provided. One of them being the **emphasis on interactivity being the key driver of visitor engagement.** Multiple authors highlight how tactile interaction, audio-augmented artefacts and participatory exhibition designs encourage deeper cognitive and emotional involvement with museum artefacts.

Additionally, there are other emerging patterns like **the use of identity-based narratives and showcases related to themes of health, memory and lived experiences to make the exhibitions more relatable to visitors.**

However, a noticeable gap across this is the fact that most authors say that **interactivity improves engagement, but sensitivity may not be considered.**

More specifically, there is a limited discussion of how interactive systems may affect different **identity groups** unevenly particularly, when it comes to dealing with **emotionally sensitive artefacts or narratives**.

Suggesting that while interaction is widely promoted to be the main reason for engagement within visitors in exhibitions, the **psychological safety is not guaranteed and remained under-explored.**



Youngblood et al. (2020) explain in **Rethinking Users** that "synthesis is not about summarising data but about transforming it into insight—revealing what is happening, for whom, and why it matters" (pg. 78)

Why these Methods?

To **critically enhance** my research, **I explored a range of methodologies** that supported different stages of the project and guided the development of my investigation. One of the primary methods used was **non-participant observation, a form of primary research informed by Stickdorn et al. in This is Service Design Doing (2018).**

The text emphasises that **practical and experience-based approaches to research can generate deeper insights than relying solely on secondary sources.** By observing participation and behaviour within real-world contexts, I was able to develop a **more critical understanding of visitor interaction and engagement within the exhibition space.**

Alongside observing other visitors, I also conducted an **autoethnographic walkthrough, documenting my own experiences, reflections, and emotional responses within the exhibition.**

This allowed me to analyse the space from both an **observational and personal perspective,** creating a broader understanding of the visitor experience.

Based on these observations, a **variety of visual and analytical methods were developed, including spatial relationship diagrams, system maps, and gestural notations.** These methods helped visualise behaviours, movement, interaction, and relationships within the exhibition environment.

This process eventually led to the creation of a research wall, which became a **central tool for organising and synthesising my findings**. The research wall allowed me to **identify patterns, themes, tensions, gaps, and connections across my observations, readings, and secondary research, ultimately supporting the refinement of my research direction and questions.**

These methodologies were essential to developing a deeper understanding of the topic, as secondary research alone was not sufficient. Instead, secondary sources acted as a foundational starting point, while primary research enabled a more focused and critically informed investigation.

This approach also reflects **Booth's The Craft of Research (2016), particularly the idea that meaningful research emerges through refining broad interests into focused and investigable questions.**



Autoethnographic Walkthrough

Initial Emotional Response

Upon entering the exhibition, **I experienced immediate visual intrigue**, more specifically, in response to the **transparent woman (plastic wire, c.1980, restored 2006).** The illuminated skeletal and nervous system, activated through interactive buttons, created a sense of curiosity, accessibility and a new way to learn.

What Drew Me In

The **high level of interactivity across multiple displays immediately distinguished the exhibition from traditional museum formats**. Physical touch, sound interaction, and digital immersion appeared central to the experience.

Moments of Slowing Down

• The **jukebox installation** (interactive song selection related to illness, epidemics, and health narratives globally).

- Hearing multiple genres including some I **culturally identity and relate myself** too sparked a moment of happiness and curiosity.

• The **digital animation *White*** (8 min 28 sec loop), where immersive sound design created a sense of environmental presence.

Moments of Confusion / Friction

• The **astronaut-like sculpture** carrying multiple objects lacked immediate contextual framing. Unlike other artefacts, it did not provide explanatory information at the same density, creating confusion.

• **Prosthetic artefacts and medical devices (leg, breast, dyslexia glasses, hearing aids)** created cognitive pause due to their proximity to lived bodily realities.

Emotional Shift

• The **“Blood Objects”** display produced a significant emotional shift after learning that the objects were created using **HIV-positive blood**. The emotional impact appeared to intensify after reading contextual information.

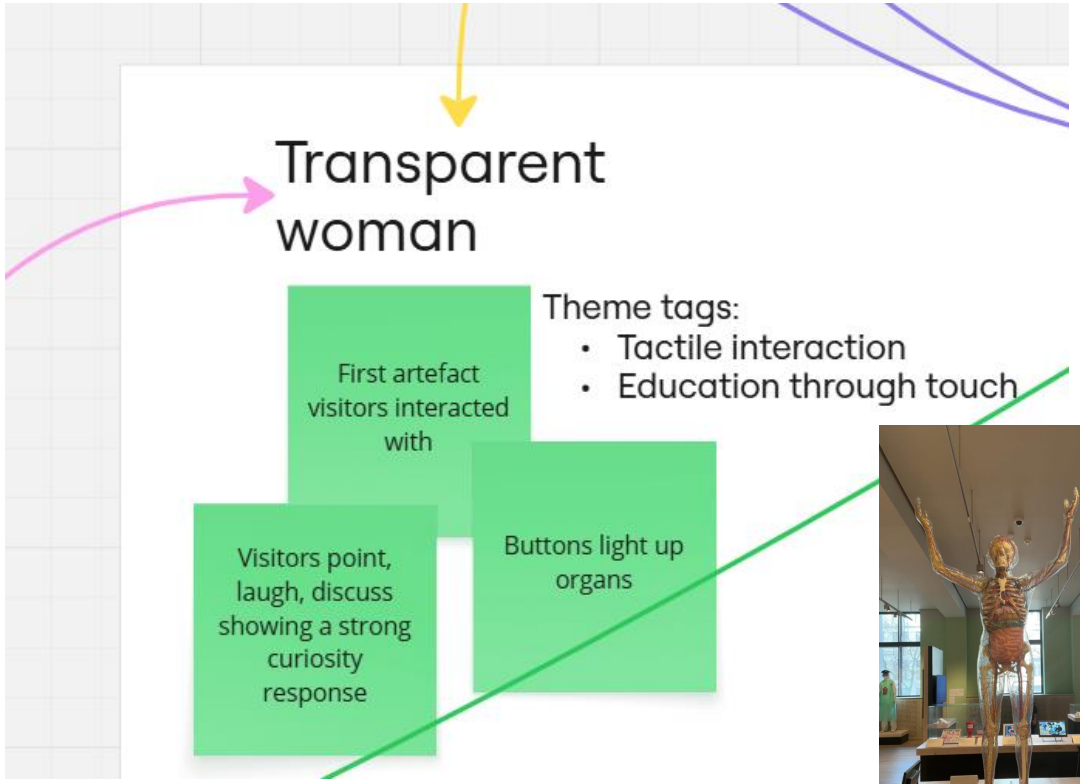
IDEO’s Field Guide to Human-Centred Design (2015) states that “You cannot design for someone unless you understand their perspective” (pg. 23).

This informed my autoethnographic approach by encouraging me to experience the exhibition personally to better understand how emotion, sensory interaction, and interpretation shaped visitor experience.

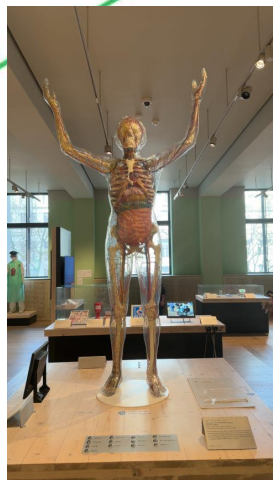
Transparent Woman

Transparent Woman (Anatomical figure)

- First artefact **many visitors approached.**
- Immediate engagement after reading **“please touch.”**
- Frequent **pointing, laughter, close observation**.
- Visitors **pressed multiple buttons in sequence**.
- In group settings, **members took turns interacting**.
- Longest recorded engagement **(group of 5): 2 min 34 secs (one individual remained for full duration).**



Screenshot taken from my **Research wall**.



Notable pattern:

Explicit permission to touch resulted in immediate tactile engagement without hesitation.

The Jukebox

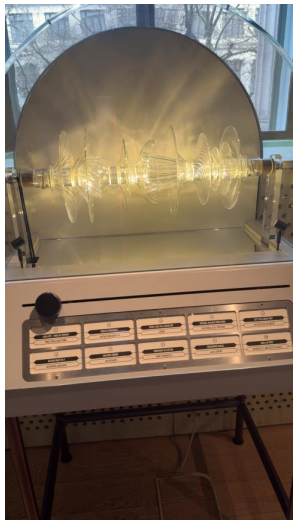
Jukebox Installation

• Visitors explored **multiple song options before selecting their preferred track.**

• Some played all **available songs briefly before moving on**.

• Others **selected one and listened for extended time (up to 1 min 7 secs).**

• One visitor did not interact initially **(11 secs observation), left, returned briefly (3 secs),** then exited.





Screenshot taken from my **Research wall**. There's more information on this specific artefact like **drawings**, due to this becoming my **focused point** on my project.

Will be explored in the later slides.



Notable pattern:

Interaction created experimentation behaviour rather than passive consumption.

Emotionally Sensitive Displays

Behavioural Observations

- Some visitors **skimmed large text panels (approx. 7–8 secs)**.
- Others **leaned forward, stood very close to text, and read extensively.**
- Groups often designated one **"knowledgeable" member** who **verbally explained content** to others.
- **No whispering observed**; tone remained conversational.
- Some visitors **pointed at specific textual segments while discussing**.

Blood Objects

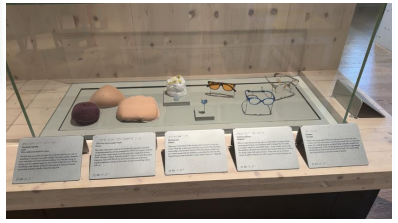


Artefacts made from HIV-positive blood

Theme tags:
- Emotional sensitivity
- Identity and health narratives

Engagement increases after reading text/narratvie





Screenshot taken from my **Research wall**.

Notable pattern:

Engagement depth varied significantly depending on interest, prior knowledge, and group dynamics.

Dwell Time Tracking

Artefact	Visitor A	Visitor B	Notes
Prosthetics	28 secs	7 sec	Significant variation
Accessibility icon display	8 secs	12 sec	Minimal engagement
Jukebox	1 min, 7 sec	43 sec	Interaction driven
Transparent Woman	1 min, 20 sec	51 sec	Cognitive engagement
White (digital animation)	1 min, 35 sec	3 min, 10 sec	Immersive attention

Notable pattern:

Interactive or immersive artefacts consistently produced longer dwell times than static mediums like the medical devices.

Environmental and System Analysis

Accessibility

- **Audio description station** observed in use.
- **Large print, transcripts, magnifiers, BSL interpretation** available.
- **Accessible wheelchair** space.
- **Seating integrated** near most artefacts.

Accessibility appears embedded rather than appended.

Signage

- **Clear, concise, contextual**.
- Touch permission signage **visibly reduced hesitation**.





Lighting + Sound

- No reported reading difficulty.
- Sound not disruptive; **some installations immersive**



Interaction Typology Map + Insight Directions

Type	Artefact example	Engagement Style
Tactile	Transparent woman	Press, touch, explore
Auditory + choice	Jukebox	Select, compare
Immersive visual + audio	The White	Sit, watch
Static Informational	Prosthetics	Read, skim

Youngblood et al. (2020) explain in **Rethinking Users** that "synthesis is not about summarising data but about transforming it into insight—revealing what is happening, for whom, and why it matters" (pg. 78)

Tactile Permission

Explicit invitation to touch like a signage stating a visitor is allowed to touch, appears to reduce hesitation and in return, encourages embodied engagement with medical artefacts.

Agency and Control

Interactive installations can generate exploratory behaviour, which also suggests that visitor-controlled systems like the Jukebox can increase dwell time and experimentation.

Embodiment vs Observation

Artefacts that require **physical participation to view results, appear to produce longer and more social engagement** rather than static displays like the prosthetics.

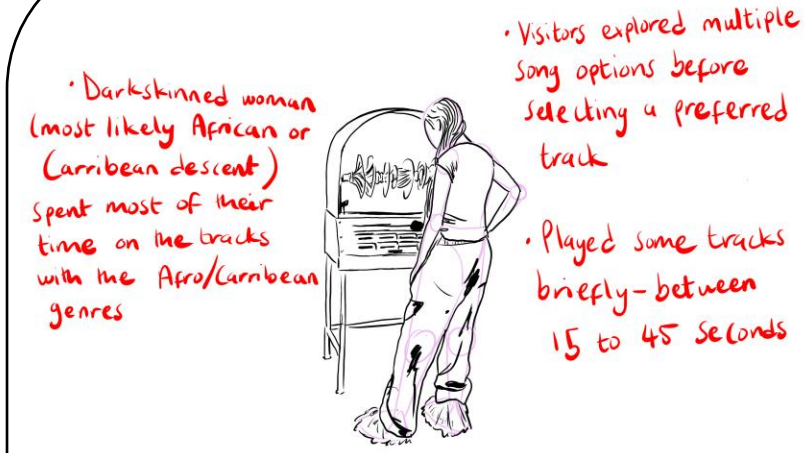
Gestural Notations



Multisensory environments involving artefacts like sound augmentation, increase the emotional engagement and focus of visitors.

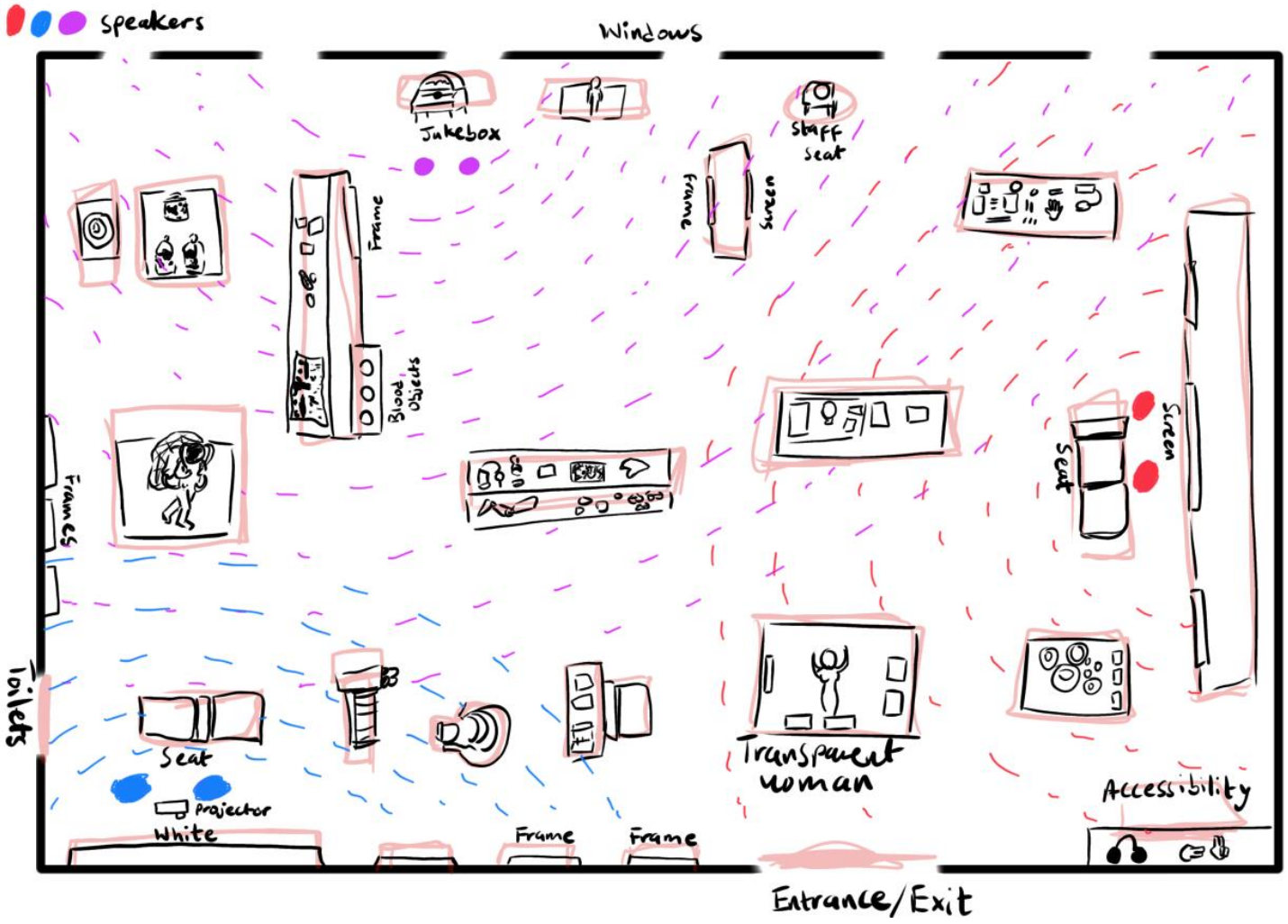
Identity-based audio content relates to visitors; it can bring groups together because they relate to it in their own experiences.





Visitors engage with identity-based audio content selectively because not only do they seek personal relevance but once they do, it is related to their personal experiences and sense of identity.

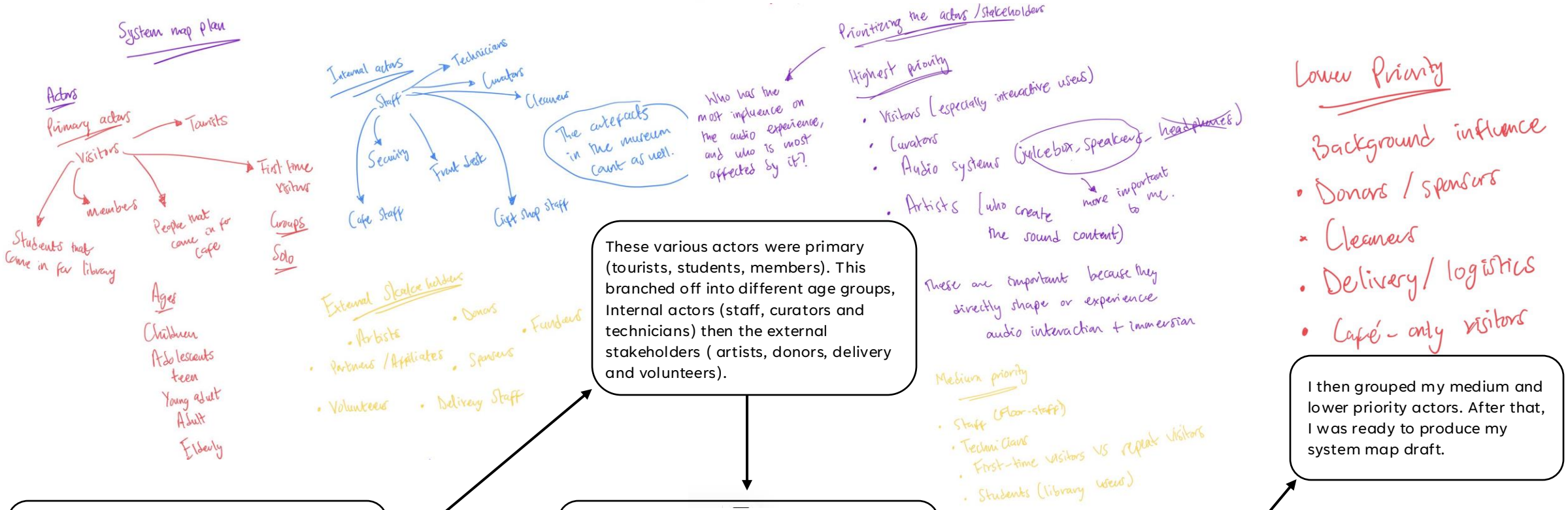
Spatial Relations Diagram



I created a bird's eye view of the exhibition including what my focus is being immersive sound artefacts within the exhibition. I have highlighted where every speaker is for each audio-augmented artefact and displaying how far the sound travels from each of them.

As you can see, the Jukebox has the largest radius of sound. When I walked into the exhibit, the first sound augmented artefact I heard/noticed was the Jukebox. This brings in and attracts visitors due to their wondering of where said sound is coming from.

Systems Map Plan



These various actors were primary (tourists, students, members). This branched off into different age groups, Internal actors (staff, curators and technicians) then the external stakeholders (artists, donors, delivery and volunteers).

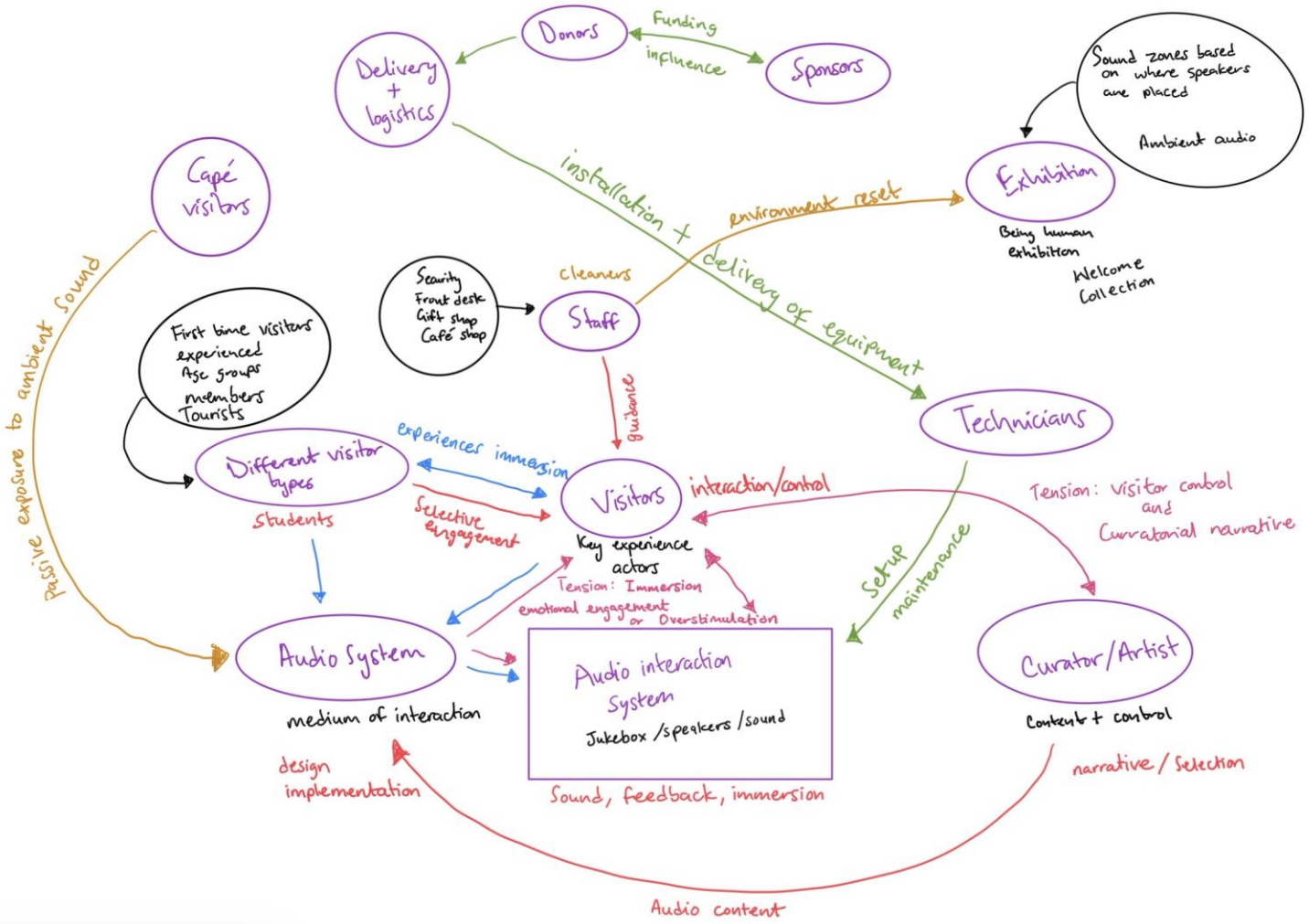
The planning for my system map shows my consideration in thinking about who and what is involved. Thinking about the various types of actors.

Once I listed each actor/party involved within the exhibition's system, I grouped them up based on priority. Highest priority, medium and then low priority.

The highest priority actors were what directly shaped the experience in the exhibition. More specifically, the audio interaction + immersion experience. This included visitors, curators, audio systems and artists creating the sound content.

I then grouped my medium and lower priority actors. After that, I was ready to produce my system map draft.

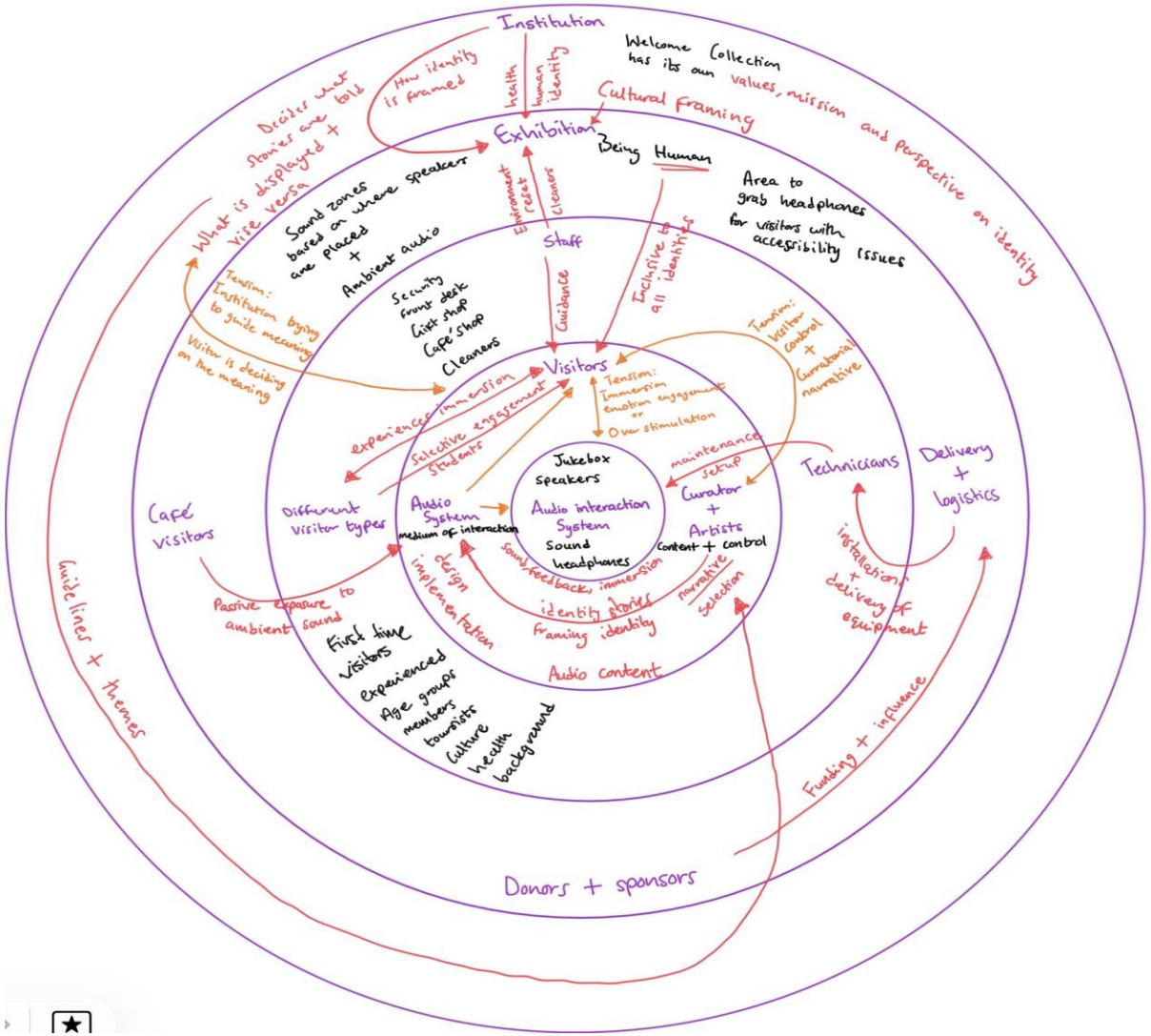
Systems Map Draft 1



This is the first draft of my systems map on the Being human exhibition. Including all my researched and planned information. Each actor, within their priorities and how they all link to each other.

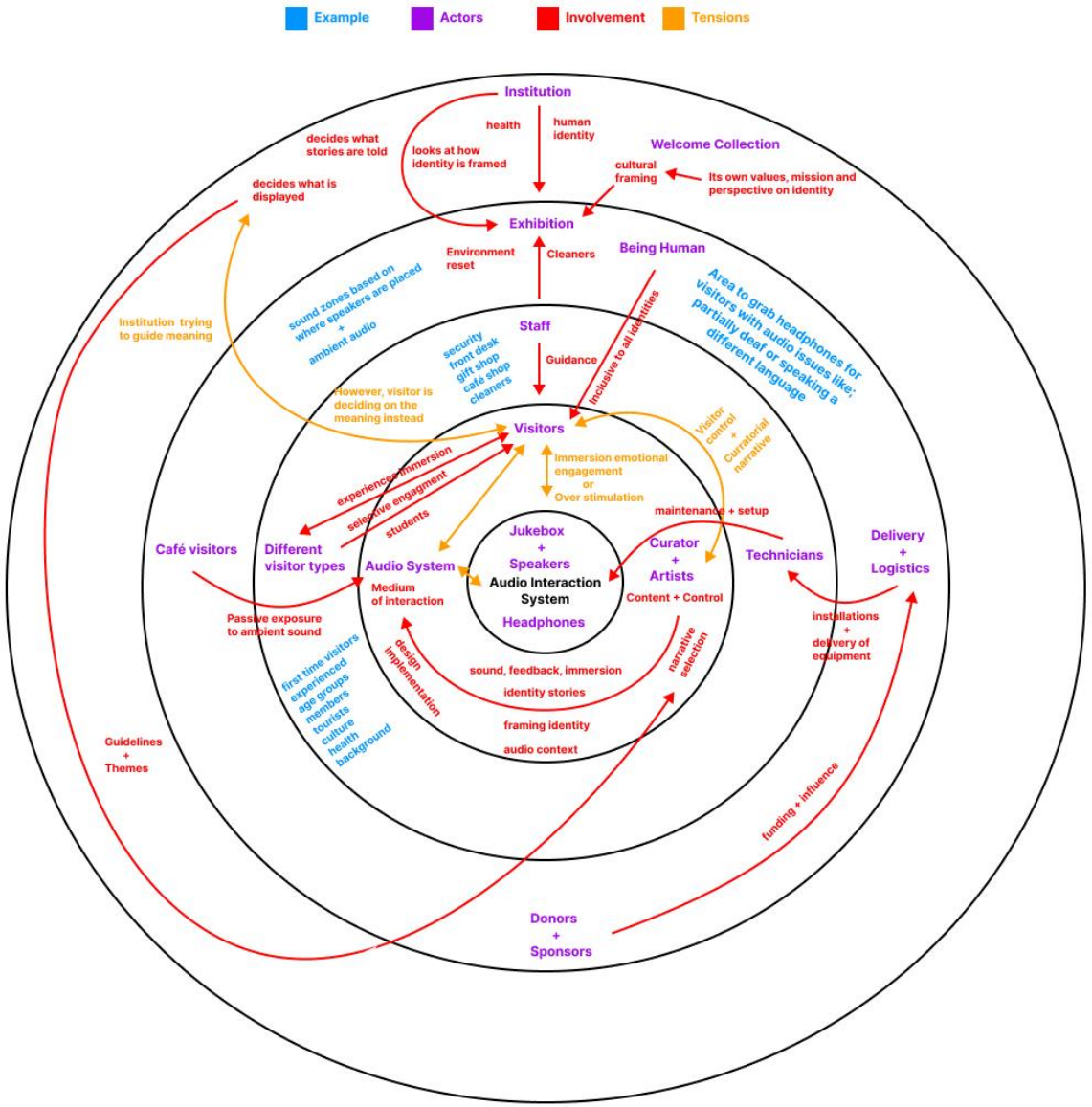
However, the way this system map was designed appeared very rushed, crowded and scattered. It looked more like a mind map or brainstorm than an actual planned out system map. This made it very hard to track and read because everything seemed all over the place creating cognitive load and confusion.

Systems Map Draft 2



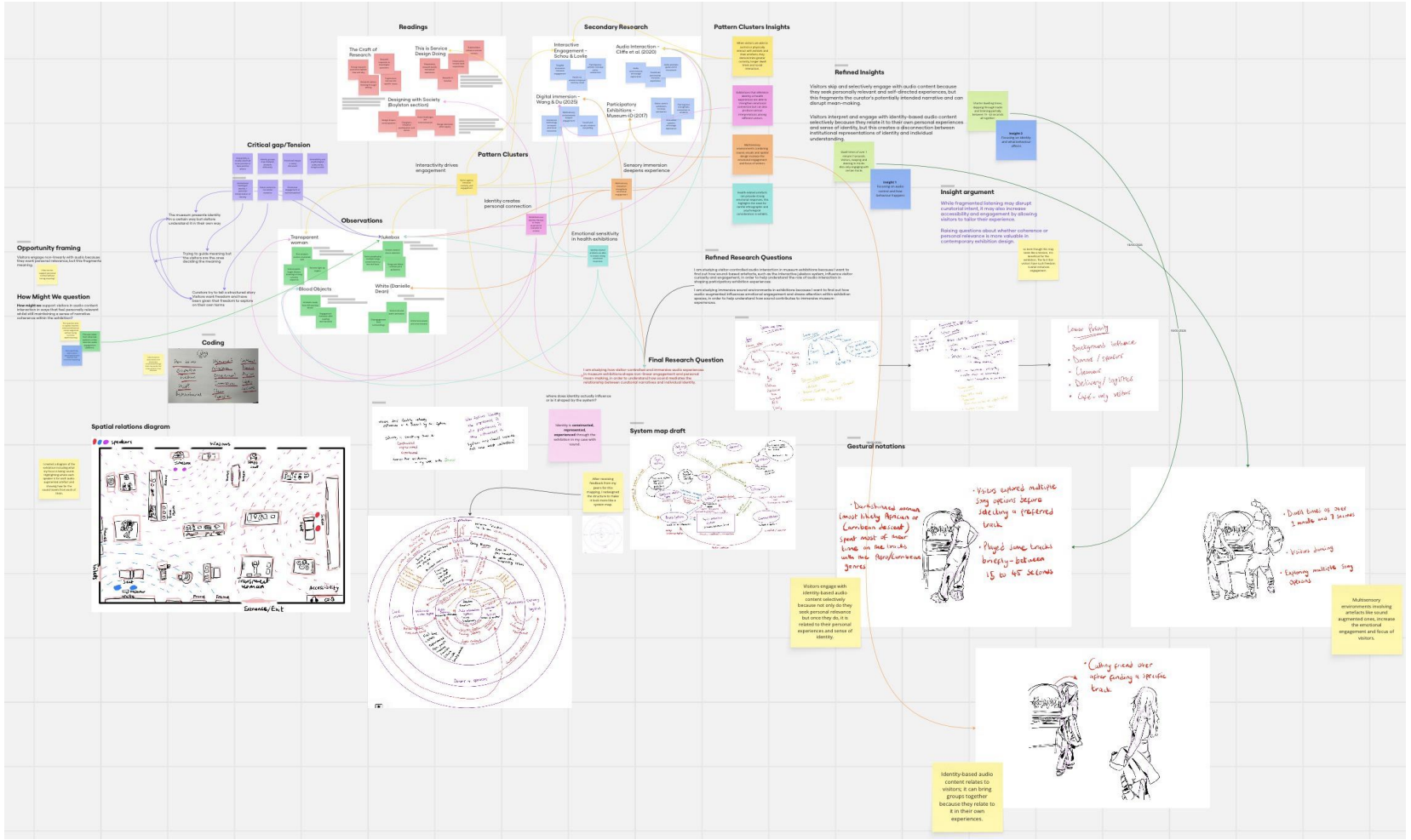
This is the second draft of my systems map on the Being human exhibition. As you can see, every problem faced with the first draft was drastically improved due to peer feedback. This made my system map a lot easier to read and track, there is clarity and understanding; this time appearing as a proper system map rather than a crowded/dense brainstorm.

Finalised Systems Map



My final systems map, has been improved. Easier to read and understand due to the enhanced legibility and specifically, the colour coding. This allows tracking of the systems map smoother and with less cognitive load.

Research Wall



This is a large screenshot of my whole research wall. Consisting of a variety of information accumalted through research and practice. Readings, secondary research, observations and all drawings/diagrams.

As you can see these were all linked and connected to each other based on their relevance with each other. Some secondary sources linking and relating to readings and obversations made.

Branching off them, patterns emerged, critical gaps/tensions and insights. These all also linking with each other in support to producing my final research question and my How Might We question.

Research Wall link: https://miro.com/app/board/uXjVGzSop8s=/?share_link_id=264742891030

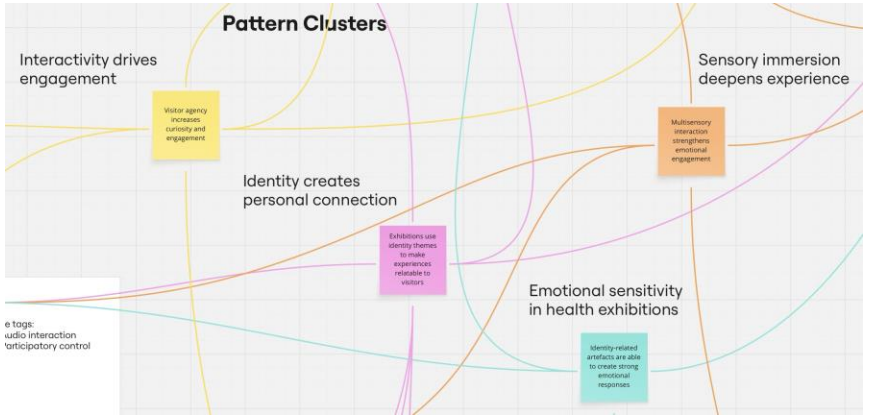
Pattern Recognition

From my Research wall, **patterns begun to emerge, this would be useful to approach because it would be a great foundation to produce my emerging insights and refinement of my final research question.**

To begin, I made a **coding of words** based on my observations and information accumulated through them. These keywords made up what I had observed and were the first steps of my patterns:

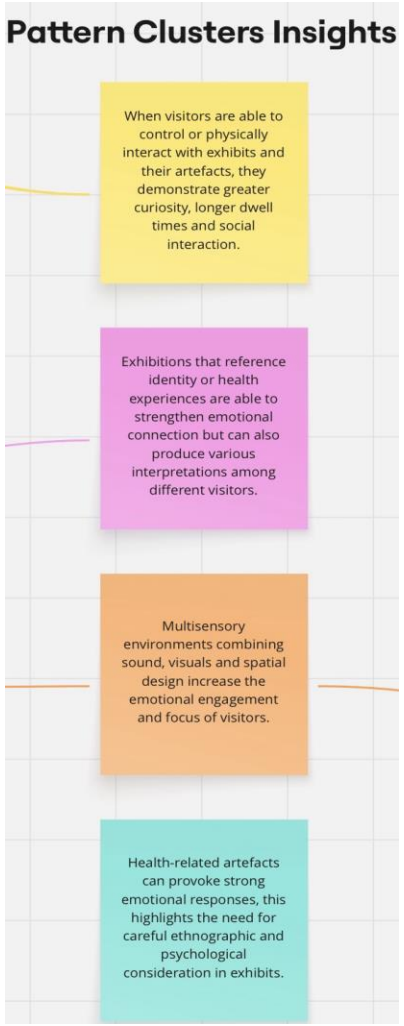
- Slowdown
- Confusion
- Friction
- Shift
- Behavioural
- Contextual
- Tactile
- Auditory
- Static
- Hesitate
- Skimmed
- Minimal
- Engagement
- Immersive
- Clear
- Concise

Coding helped transform **observations into broader themes and interpretive insights**. Through analysis, **patterns within the data were identified**, while synthesis enabled these patterns to be interpreted and connected to wider meanings and research questions.



This led to my pattern clusters; each grouped into their own sections. **Enagement, personal connection, emotional sensitivity and sensory immersion.** I made a brief description on each as you can see on the sticky notes.

However, I went more in depth when it got to the pattern cluster insights. Asking myself for example; **“yeah, interactivity drives engagement but why”, “what about the health-related artefacts, interpretation can’t be the same with all visitors right?”** Leading to my pattern cluster insights.



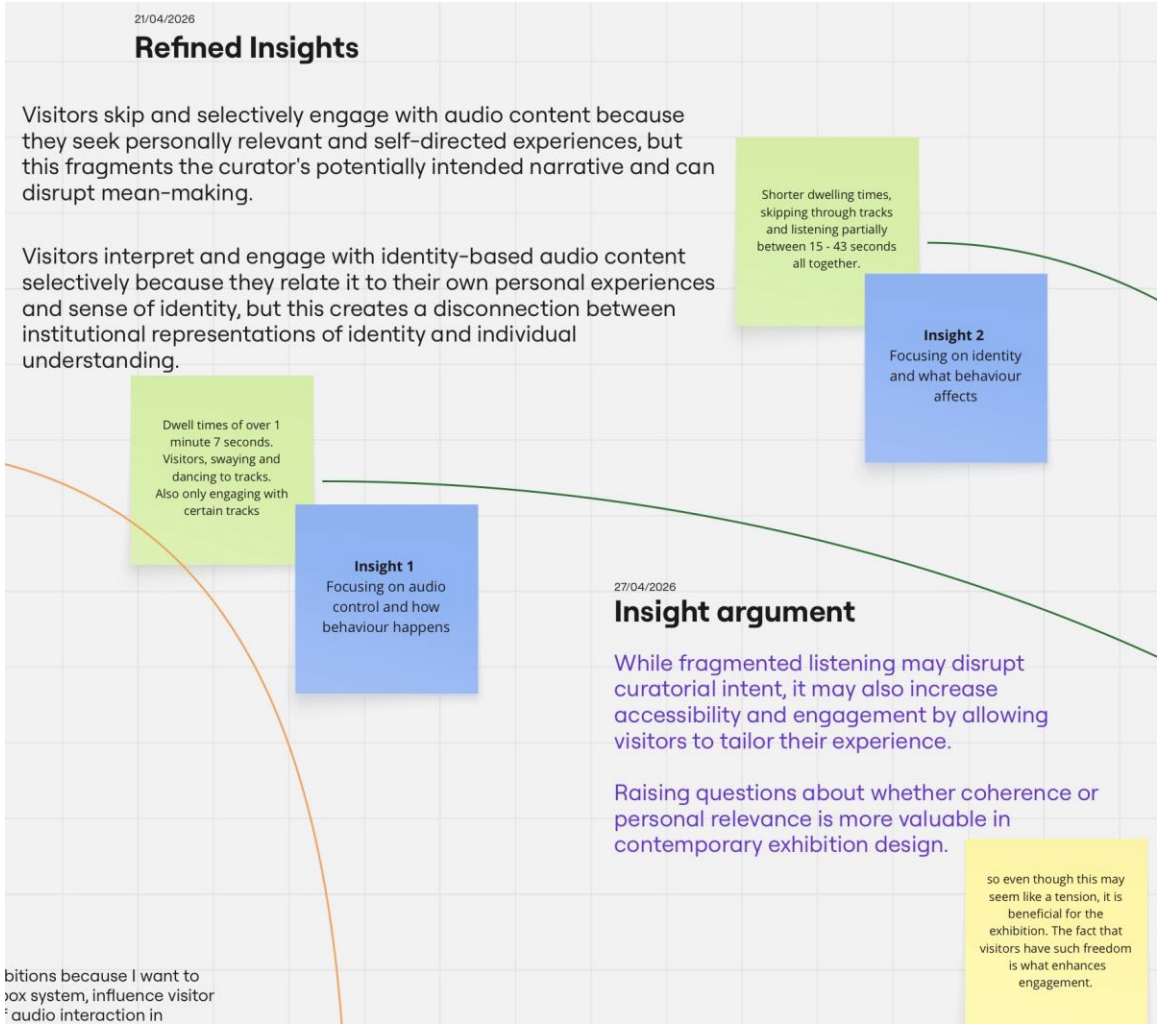
Refined Pattern Insights + Arguement

By applying knowledge from the **Craft of Research by Booth, et al. (2016) – narrowing topics to specific questions and from feedback from peers and tutors.** I was aware that I needed to refine my insights. It was okay to have multiple insights however, my problem was the fact that **I had four various statements, and some had different concerns/subjects than others**. This meant that I needed to refine and focus on one subject matter which is what I done.

To show critical analysis/thinking I made sure to not only write an insight but include a drawback which could lead into an emerging question in that insight. For example, I stated; "Visitors skip and selectively engage with audio content because they seek personally relevant and self-directed experiences, **but this fragments the curator's potentially intended narrative and can disrupt mean-making**". I am not necessarily focusing on the second part being the drawback alone, insetad I state it to display my in-depth analysis on the whole picture.

When writing my insights, I also made sure to include evidence to support them. **Evidence based on my observations, and linking them with arrows, connectining to things like my drawings, clusters and readings.** This is because **according to Stickdorn et al. (2018), key insights made should be built on research data and be supported by evidence.**

Just like how I added a drawback but my refined insights. I wanted to add an argument that went against the general idea of my insights. **Explaining how these insights can be beneficial but also be a ciritical tension within the exhibiton and visitors.**



Refined Research Questions – Final Question

In Page 3 of my process book, I produced five different research questions that briefly talked about:

- **How interactive exhibition design influence visitor engagement.**
- **How exhibitions use identity-related narratives.**
- **How sound, visuals and spatial immersion influence visitor engagement.**
- **How sound environments and visitor-controlled audio systems influence movement, attention and engagement.**
- **How interactive displays can both engage visitors and potentially trigger emotional discomfort.**

As time went on, my research was being refined through writing and each time, my topic was being narrowed into my focus. Looking at the **Craft of Research by Booth, et al. (2016)** I was able to condense my ideas into a smaller subject.

Initially, choosing between three themes; **identity, health and trust**. I looked at each theme critically, through research and observations and linked it to my focus. Confirming to me that from my insights, **identity has been represented, constructed and experienced through the exhibition with sound**.

Which my question was then based on **controlled audio-augmented artefacts and their influence on visitor engagement.**





"Analysis is about breaking complex problems down into smaller pieces. Synthesis is about building up a coherent understanding of what the research means." — **Stickdorn, Hormess, Lawrence & Schneider, This Is Service Design Doing (2018)**

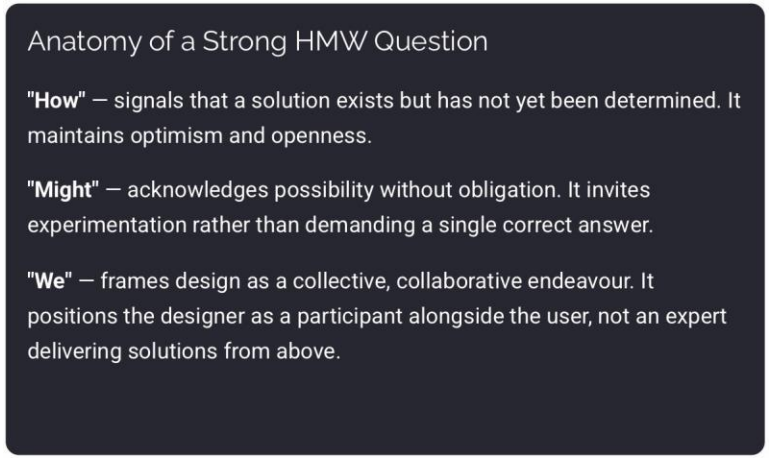
Opportunity Framing + How Might We?

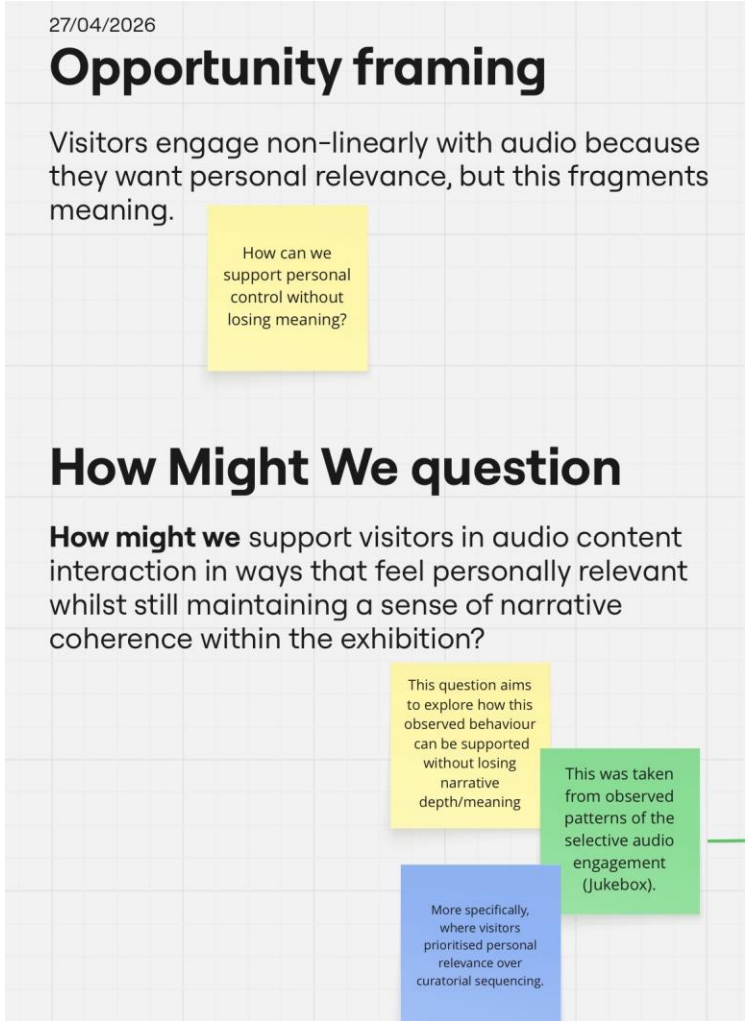
I learned that the **How Might We question is one of the most widely used tools in human-centred design practice. It has been developed and even popularised by IDEO, Google, Luna Institute and design schools worldwide.**

My point is that the How Might We format provides a **structure that is able to translate an insight into a design opportunity you are able to act on**.

As I formed my question, I made sure to **not make my question too narrow but also not make it too broad**, I also made sure I did not assume the solution before understanding the problem. **This meant I had to break down my problem into various parts.**

Reviewing my codes, naming the pattern, interpreting the pattern, writing my insight and then framing my HMW question. Making sure I am specific and bold in my interpretations moving away from what I saw to what it means. This led to the understanding of my insights and supported in the framing of my HMW question as I followed its anatomy.



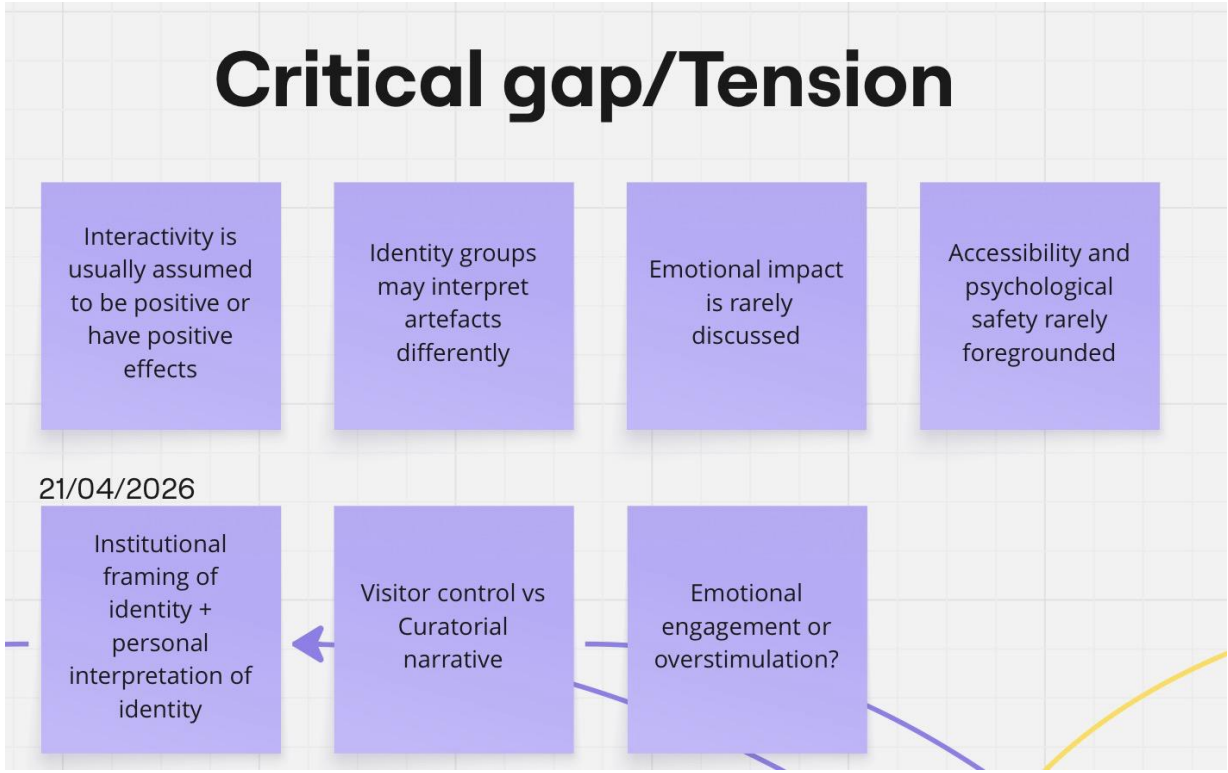


Screenshot taken from a slide presentation created by my tutors.

Critical gap/Tension

To highlight critical thinking and display understanding, **I investigated what critical gaps/tensions may emerge from the patterns within.** Focusing on visitors and how the exhibition presents identity that they can relate to through sound.

I realise that even though the exhibiton can present identity. It is presented in a **certain way however; visitors understand this in their own way,** and they decide what something is presented means. This is because, as much as the exhibition **tries to guide meaning, visitors have more freedom and agency** with the artefacts to explore what ever structured story the curators are trying to tell more like from Jukebox.



Final Insight Direction

As my research has evolved, I have moved beyond **”audio increases engagement” to “audio interaction shapes personal and identity-based meaning making”.**

Visitor-controlled and immersive audio experiences within exhibition spaces have done more than increasing the engagement of visitors. **They shape how visitors emotionally connect, interpret meaning and relate exhition content to their own identities and lived experiences**. My observations on interactions specifically with the **jukebox** have revealed so.

I discovered that visitors often selected audio based on **personal interest, cultural familiarity or emotional relevance rather than following the curated sequence of the tracks**. I noticed in some cases, **visitors were calling others over to listen to specific tracks**, which suggests that sound can bring **shared social and emotional experiences** within the exhibition space together.

These behaviours observed, indicate that **audio interaction functions as a form of personal and cultural mean-making**, where instead of following the order of the tracks set, they navigate through the content based on what they feel close with, whether it is their **memories, identities or experiences**. This reflects theories of **participatory culture, Jenkins, Museum-iD, FIWebteam (2017).** Talking about audiences shaping their own experiences positioning meaning as something that is individually interpreted and socially influenced.

At the same time, **tension is created between visitor agency and curatorial intent.** Linking to my **critical gaps/tensions, HMW question, refined insights, argument and final research question**. This is because while curators may design immersive audio experiences to communicate structured narratives around identity and human experience, **visitor-controlled systems gives users the agency to engage selectively which potentially fragments the intended narrative**. For example; the jukebox tracks may have been put in a certain order, **alphabetical, rhythmic or chronological order in time**. However as stated, it is fragmented by visitors.

On the other hand, this selective engagement may also deepen emotional resonance by enabling visitors to **prioritise content that feels personally or culturally relevant.**

As a visitor, if I could not control the tracks and play what I wanted, I doubt I would spend time waiting for all 12-16 tracks to play until I heard the song I wanted to listen to again. Which does **affect dwell times drastically linking directly to engagement level.**

Therefore, this project explores **how immersive and visitor-controlled audio experiences influence not only the engagement and attention, but also how individuals are able to interpret identity, construct a person meaning and then form emotional or social connections linking to the sound within exhibition spaces.**

Principle 03 – Futures Thinking

This project connects to futures thinking because it **explores how immersive and visitor-controlled audio experiences may shape exhibitions that are more interactive and participatory as the Being Human exhibition.**

Through observing visitor behaviour, I understood how sound can not only influence engagement and attention, but **reveals identity, emotional connection and shared experiences between visitors**.

Through research, it is suggested that future exhibition experiences may become **increasingly personalised and self-directed, allowing visitors to navigate content through their own cultural backgrounds, interests and lived experiences**. Subsequently, questions are raised, questions around **accessibility, narrative control and inclusivity within exhibition design.**

This project does wonderfully, by going **beyond the focus of usability and considering how interactive systems may contribute to more meaningful, human-centred and emotionally aware experiences in future museum/exhibiton spaces.**



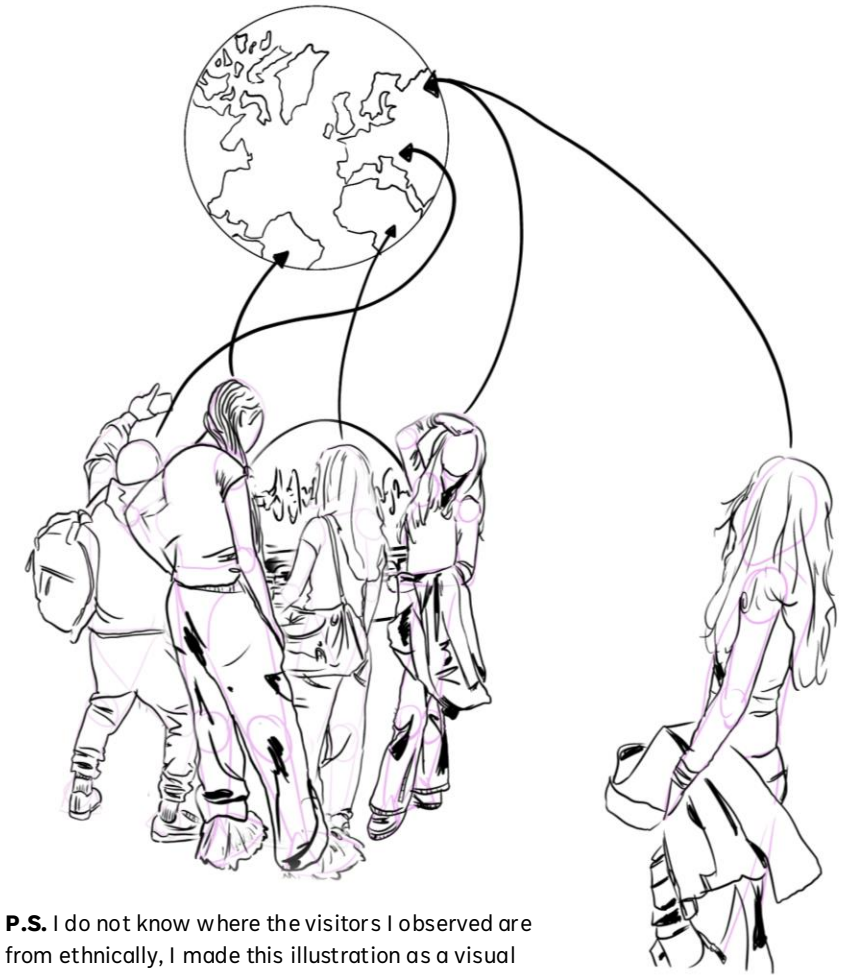
Principle 04 – Human Equity and Justice

This principle became highly relevant throughout my research because **visitor interaction within the exhibition was strongly shaped and influenced by personal identity (chosen theme), cultural familiarity and emotional relevance.**

As I observed, I noticed visitors would often **engage selectively with audio content that reflected their own experiences or backgrounds**. For example, some visitors appearing to be drawn toward culturally familiar music or narratives like afro beats and carribean music, whereas others **shared certain tracks socially with other visitors within their groups.**

This highlighted how engagement is not **experienced equally or universally**. Instead, engagement is **influenced by individual identity, memory and lived experience.** This as a result, led to me questioning how exhibitions can frame identity institutionally (as a whole), while visitors **reinterpret these curated narratives through personal mean-making.**

Therefore, reflecting an equity-centred perspective by recognising that users do not all engage with systems the same but differently depending on their **social, cultural and emotional contexts/background**.



P.S. I do not know where the visitors I observed are from ethnically, I made this illustration as a visual description for what this slide talks about.

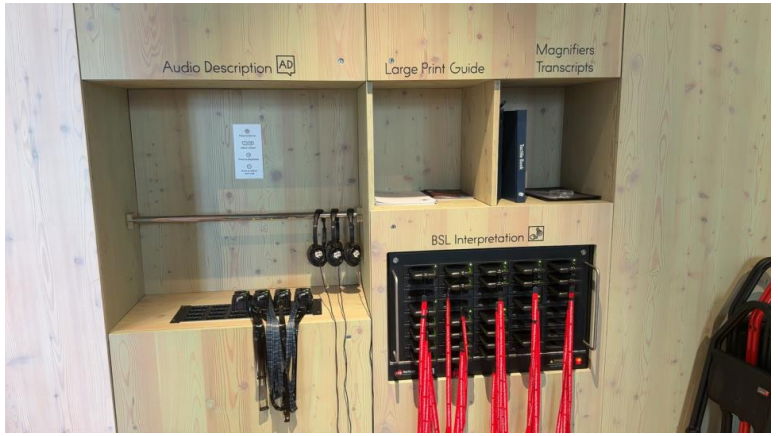
Principle 05 – Activism and Adovcacy

Another principle this project connects to is **activisim and advocacy through its focus on participation, representation and inclusive engagement within exhibiton spaces**.

From the investigations made on how visitors **emotionally and culturally connect with sound-based experiences**, the research conducted has highlighted the importance of d**esigning environments that allow individuals to feel represented, included and emotionally connected** like; the jukebox having a variety of genres for most cultures, and the accessibility box for visitors hard of hearing or speaking a different language.

The project encourages reflection on whose identities and experiences are prioritised within exhibitions and how interactive systems can reinforce or challenge dominant narratives.

Through my observations, more specifically, on **visitor agency and selective interaction**, I began to understand how participatory experiences can support more **inclusive and most of the time are socially aware** of the various types of visitors they may encounter. Therefore, including **multiple forms of engagement, where visiotrs are not passive consumers of information but instead, active contributors to the mean-making of the exhibition**.







Reflective Connection

Principles 03 and 04 became increasingly relevant throughout the process of my project. From **focusing broadly on interaction and immersion to now understanding how audio experiences influence identity, emotional connection and personal meaning-making within exhibitions.**

From my observations on visitor behaviour, I realised that engagement was not only shaped by design or usability, but it was beyond that, reaching reasons like **visitors' cultural backgrounds, memories, emotions and lived experiences.**

Directly connecting to my **final insight direction (Pg. 34 of my process book)**. Exploring how visitor-controlled and immersive audio experiences are what **allows visitors to interpret exhibition content through their own identities and personal relevance.**

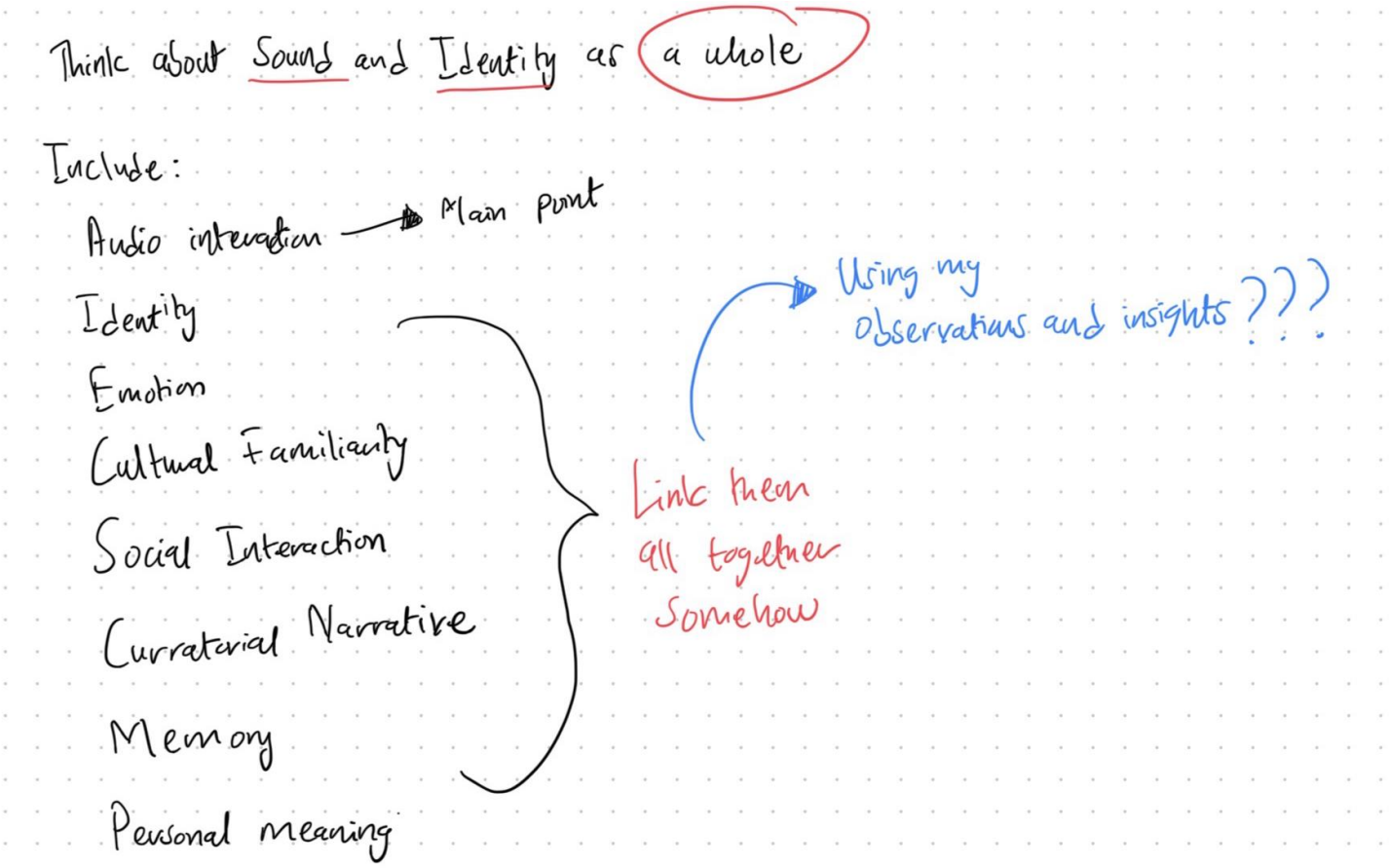
Principle 03(Futures Thinking) supported me when it came to considering how future exhibition environments may become increasingly participatory, emotionally responsive and socially away/inclusive through sound-based interaction or maybe interaction in an exhibition in general.

At the same time, **Principle 04 (Human Equity and Justice)** encouraged me to critically reflect on how identity is represented as a whole, and how visitors selectively engage with content that resonates directly with their own life experiences.

Altogether, **these principles aided me when it came to understanding that exhibition spaces and what consists of them are not experienced universally but meaning is actively constructed by every visitor individually.** This can be through emotional, cultural and social interpretation because music is a very power tool that holds all.

In return, my project was shifted away from analysing engagement and moved onto examining how **immersive audio can shape participation, belonging and an identity driven experience within exhibition spaces**.

Visual Artefact Plan



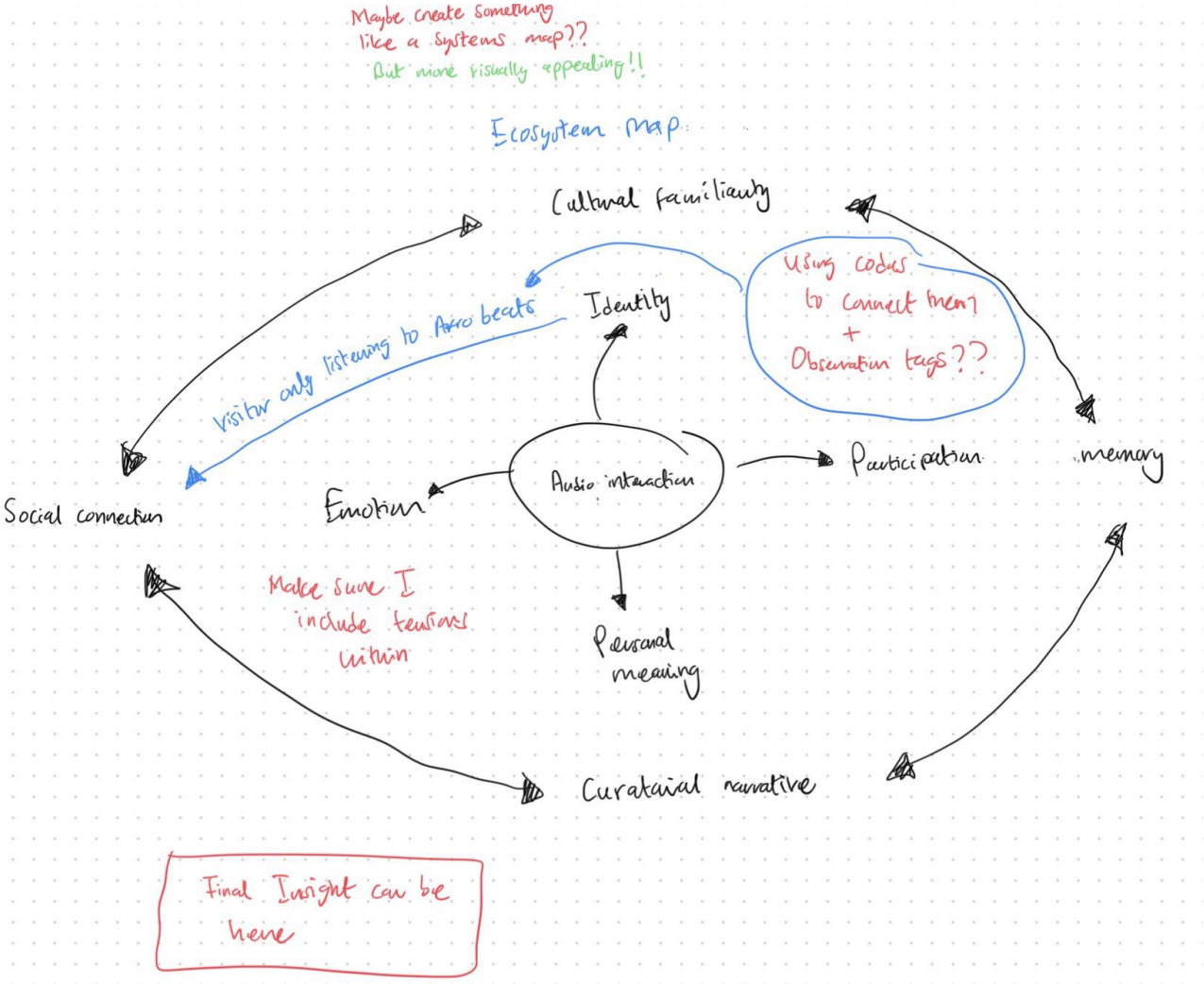
Being in **the Discover + Define** part of the project, I needed to make sure my artefact did **not look like a final design solution but just something that communicates my findings, not solve them.**

This planning was a quick brainstorm on what needs to be included before I start drafting on the design.

Listing the themes and topics my visual artefact should contain like my **observations, insights, identity and personal meaning from visitors.**

Audio interaction being my main point however, branching off into **how sound and identity link with each other.**

Visual Artefact Initial Draft

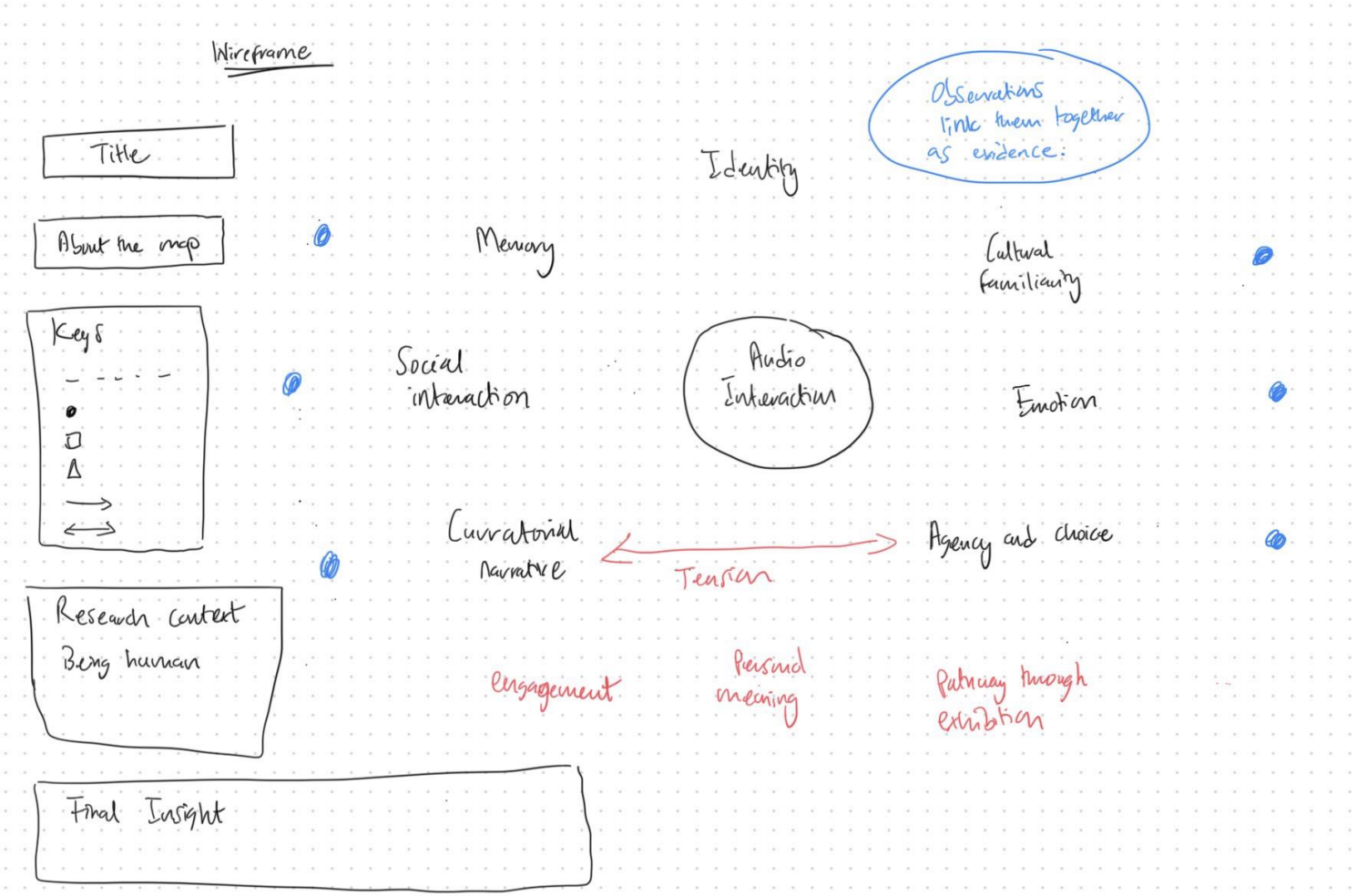


This was when I realised I can create some sort of **ecosystem map**. Where I **combine system mapping, observation, final insight, identity, emotional engagement and audio interaction and display it as a system linking together with each other**.

This would visually communicate clearly to the audience how sound and identity link together. It will end up becoming, **“This is what my research discovered” which is exactly what the Discover + Define is supposed to do.**

Using the visitor only selecting **afro-beats/Caribbean tracks becomes a visual evidence for identity influencing audio selection, which in return influences meaning-making.**

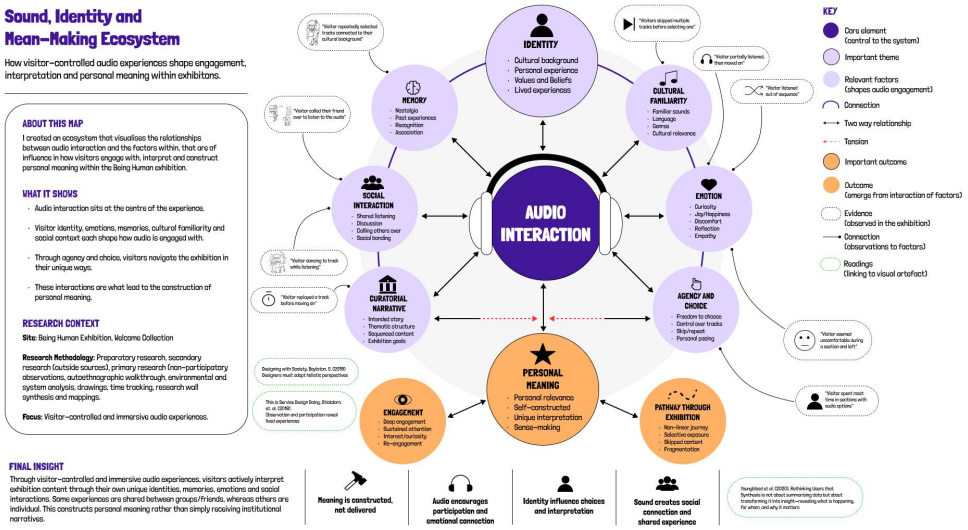
Visual Artefact Final Draft



To successfully plan out my visual artefact, I had to think about its **visual hierarchy. Making sure that with all the information on my artefact, it is still breathable instead of dense and overwhelming.**

I made sure to include an area that talks about the map, keys that aid the reader in tracking and navigating throughout my ecosystem. I added blue dots around the map, which stand for the observations that links each theme together including tensions within the links as well.

Sound, Identity and Mean-Making Ecosystem



A larger image will be displayed on the next slide.

Evidence Connection

I realised that the **evidence and observations were one of my strongest assets** within my ecosystem. So instead of listing the observations vaguely, I **attached each observation/evidence to their relevant factors** to make the map feel more **research-driven** than theoretical.

Personal Meaning Outcome

I believe that this was one of the **smartest decisions I made**. Initially, I was very **focused on the jukebox as an artefact**, but I realised that I need to focus on the **exhibition as a whole and use the jukebox, sound and interfaces as a tool** in helping me find out what my true outcome is.

Visual Hierarchy

Making sure that to display something with **clarity and to allow easy access/navigation**, I needed to think about how I would present this information successfully. Adding keys to the side, was a brilliant idea because it gives the viewers an idea on **why I used certain colours, shapes and notations.**

Clear Central Focus

By making sure **'Audio Interaction'** is in the middle, it immediately tells the audience what my project is about. Still also making sure that my theme **identity** was not put behind as a side topic but instead, I **embedded into the whole ecosystem.**

Sound, Identity and Mean-Making Ecosystem

How visitor-controlled audio experiences shape engagement, interpretation and personal meaning within exhibitons.

ABOUT THIS MAP

I created an ecosystem that visualises the relationships between audio interaction and the factors within, that are of influence in how visitors engage with, interpret and construct personal meaning within the Being Human exhibition.

WHAT IT SHOWS

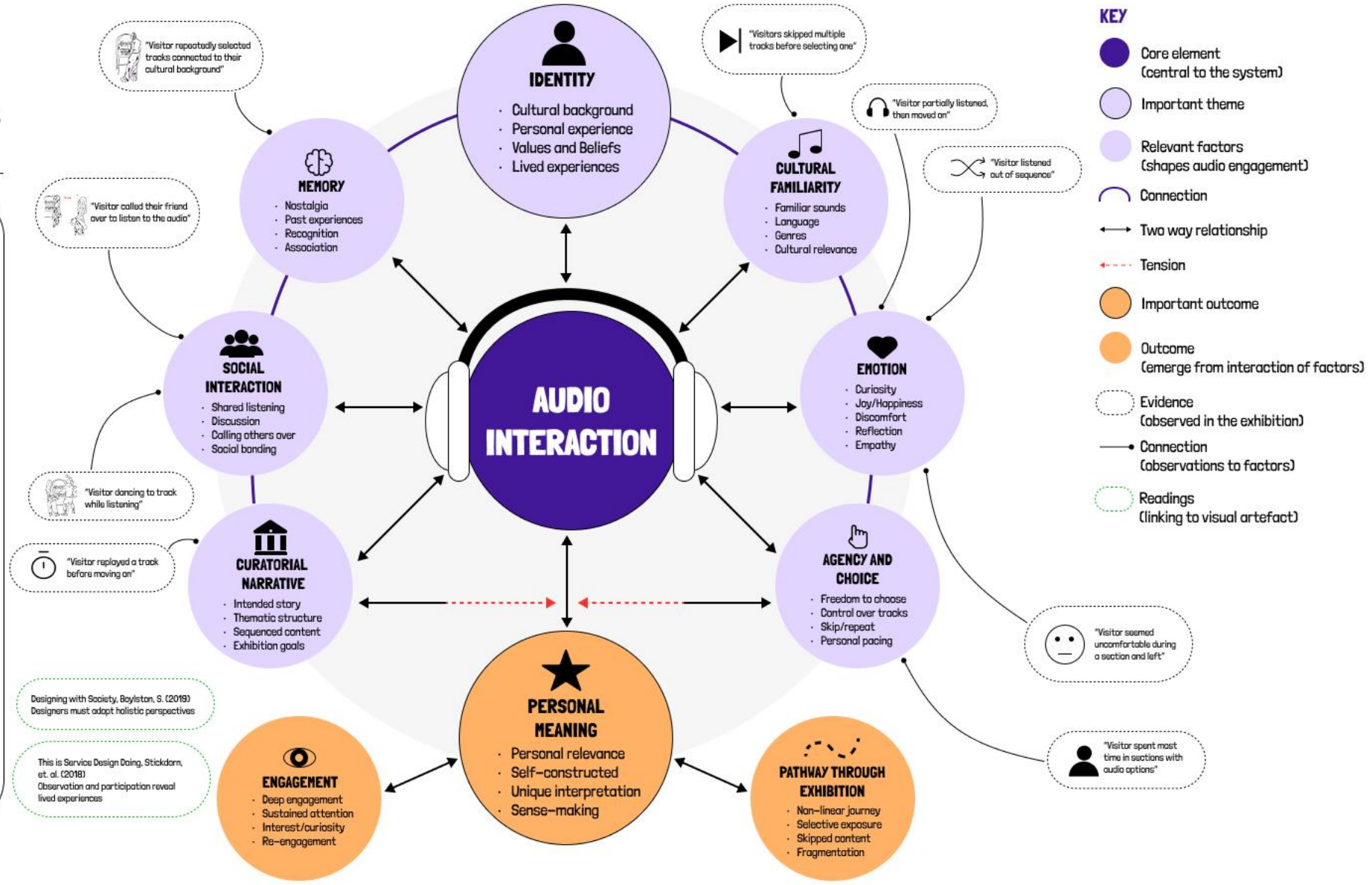
- Audio interaction sits at the centre of the experience.
- Visitor identity, emotions, memories, cultural familiarity and social context each shape how audio is engaged with.
- Through agency and choice, visitors navigate the exhibition in their unique ways.
- These interactions are what lead to the construction of personal meaning.

RESEARCH CONTEXT

Site: Being Human Exhibition, Welcome Collection

Research Methodology: Preparatory research, secondary research (outside sources), primary research (non-participatory observations, autoethnographic walkthrough, environmental and system analysis, drawings, time tracking, research wall synthesis and mappings.

Focus: Visitor-controlled and immersive audio experiences.



FINAL INSIGHT

Through visitor-controlled and immersive audio experiences, visitors actively interpret exhibition content through their own unique identities, memories, emotions and social interactions. Some experiences are shared between groups/friends, whereas others are individual. This constructs personal meaning rather than simply receiving institutional narratives.

Meaning is constructed, not delivered

Audio encourages participation and emotional connection

Identity influence choices and interpretation

Sound creates social connection and shared experience

Youngblood et al. (2020), Rethinking Users that Synthesis is not about summarising data but about transforming it into insight—revealing what is happening, for whom, and why it matters

Reflection on Learning

Throughout this project, **my understanding of visitor engagement evolved significantly**. This is because initially, **my research focused broadly on interaction, immersion and exhibition experiences within the Being Huan exhibition at the Welcome Collection.**

However, as time went on and through each research methodology; observation, synthesis and critical reflections, I began to recognise every **recurring pattern** within the exhibition based on what I saw from my observations. **These patterns surrounded subjects of audio interaction, identity, emotion and personal mean-making**. Therefore, allowing me to narrow my focus and develop a more specific line of research.

Taking from this project, the **most valuable aspect was learning how to move from data collection to synthesis. From coding observations, creating research walls and updating each time I found something new, mapping relationships and analysing tensions/gaps**. This led to the transformation of individual observations into broader insights.

Being a student, we were provided with various readings to take reference and insight from. More specifically, **Booth's The Craft of Research was the most influential, supporting me in understanding the importance of refining broad topics into focused research questions**.

Ultimately, this is what guides me to the development of my final research question and project in general because I **constantly applied** it to every obstacle I faced. I was able to move beyond pure documentation to critical analysis of them.

At the same time, **feedback played an important role throughout the project**. **Group discussions and one on one tutorials with tutors were able to highlight issues like my visual hierarchy and information structure within my process book. This led me to creating a second process book (the current one), and redesign everything to improve clarity and visual communication.**

Subsequently, **feedback from peers** were also taken into consideration whether inside or outside class. They helped me realise that **effective research communication is just as important as the research itself**; there is no point with having all this research if no one can understand what they are reading.

My final visual artefact is an **ecosystem map**, and it represents the outcome of this process. Instead of presenting audio interaction as an isolated feature, I **visually presented identity, memory, emotion, cultural familiarity, agency and social interaction as an entire interconnected system communicating how visitors engage with exhibition content**. I connected observations directly to these relevant factors which grounded my conclusions as evidence instead of assumptions.

This project demonstrated that visitors do not passively receive exhibition narratives but instead through visitor-controlled and immersive audio experiences, they **navigate through the exhibition actively interpreting content through their own identities, memories and lived experiences**.

Therefore, **constructing personal meanings rather than receiving institutional narratives.** Through this insight, my understanding of the importance user experience research, observing, analysing and synthesising human experiences has been shaped differently.

Textual Analysis

Some objects have their history imprinted on them because they are regularly in use; “we know what they are because we know what they are for” (Ahmed, 2019). Post boxes, for example, are not just decorative landmarks, but “objects with history” (Ahmed, 2019), they have been in use for generations and are determined on how a system is encountered. With confidence, we can post our letter into our post box making the post box in use; without thinking we know that it will be picked up and sent to its assigned address. Not only does this prove that familiarity provides efficient use, but it also implies that something in use does not only mean that it is currently being used.

Subsequently, to support my insight, another statement was made; “A coin that is in use is circulating but it could still be in my pocket” (Ahmed, 2019). This depicts that things can also come out and come into use. Their usage maybe suspended for future use.

This text offers a contemporary conceptual perspective in which it examines how objects acquire meaning through sustained use over time by users. Supporting the fact that history is the reason for why things feel natural or familiar because the past has been embedded in present-day and future use. While systems may acknowledge the historical continuity of such objects, its primary focus still resides in analysing how familiarity and function operate together in the present but also highlight implications for future use.

This text is based on What’s the Use? On the Uses of Use by Sara Ahmed, 2019. Published by Duke University Press; as stated earlier, it offers a contemporary conceptual perspective. Duke University Press is well known for publishing academic works in critical theory, feminists' subjects/studies and cultural analysis in return, situating Ahmed’s writing within a broader tradition of theoretical and reflective scholarship instead of a historical documentation.

Contextually, Ahmed examines how objects acquire meaning through sustained use over time, emphasising that history helps to shape why some things may feel natural or familiar since, past usages become embedded in present and future practices. Not only does her work describe objects, but it also interrogates the systems and assumptions that allow objects to function seamlessly in everyday life.

The text acknowledges historical continuity of objects however; its primary focus is to analyse how familiarity and function operate together in the present and how that complies into future practices. In doing so, Ahmed encourages the reader to reconsider ordinary objects not as passive but shaped by ongoing histories of use that we influence through how we perceive and engage with them.

Ultimately, Ahmed’s exploration of “use” exposes how everyday interactions with objects are shaped by deeper systems and histories that often seem invisible to their everyday users. Therefore, reinforcing the idea that familiarity is not accidental, but constructed through repeated practices throughout time in generations.

Case Study

Sara Ahmed's argument in What's the Use? On the Uses of Use suggests that objects acquire meaning not through inherent design, but through repeated and sustained use over time. This is evident in the statement; "they matter because they are regularly used by most as it had been for generations" (Ahmed, 2019), highlighting how familiarity can be constructed through repetition and collective practice.

In my opinion, the example of headphones demonstrates this concept clearly, since their meaning appears to have evolved through widespread and habitual use in everyday life.

Thinking of headphones, I personally associate them with personal space, focus, and control over one's environment and emotional state. However, I do not believe this understanding is inherent to the object itself. Instead, I think it has been shaped through continuous and socially shared use over time. Just as Ahmed suggests, I believe objects gain importance because they are "regularly used"(Ahmed, 2019).

This can be reflected in how headphones have become embedded within routines such as commuting, studying, or exercising in various and unique designs. Through this repetition, I think users develop an intuitive understanding of how and when to use them without needing instructions or guidance.

The idea of use sustaining across "generations" through history can also be seen, in my view, in how norms around headphone use have become widely accepted. Even first-time users may recognise that wearing headphones can communicate ideas such as 'privacy' or 'disengagement' from surrounding environments.

To me, this demonstrates how meaning is not always individually constructed in isolation but instead inherited through broader cultural practices. The statement; "we know what they are because we know what they are for"(Ahmed, 2019), feels particularly relevant here because it suggests that function is learned through exposure rather than being immediately obvious. I believe headphones are understood not just as technological devices, but also as tools for creating personal environments or temporary escape within public spaces.

I also think this concept can be applied to exhibition contexts, where audio devices such as headphones or interactive sound systems are used. Visitors entering these spaces may already bring prior experiences of personal audio with them, which could influence how they engage with sound-augmented installations. In my opinion, the familiarity of putting on headphones or interacting with audio interfaces may feel intuitive not only because of the interface design itself, but because of repeated exposure to similar systems outside the exhibition environment. This, to me, reinforces Ahmed's argument that objects do not simply present their function to users, but instead acquire meaning through ongoing interaction and cultural reinforcement.

Altogether, I believe the example of headphones illustrates how objects become meaningful through sustained use. Interactions that may initially appear natural or intuitive could be the result of accumulated practices developed over time. Ahmed's framework challenges the idea that usability is inherently understood, instead positioning it as something that emerges through constant engagement, social normalisation, and shared understanding.

Contextual Framing

Sara's Ahmed's What's the Use? On the Uses of Use and the article Creature of Habit: A Self-Report Measure of Habitual Routines and Automatic Tendencies in Everyday Life, both explore the relationship between habit, familiarity and repeated behaviour but from different perspectives. Ahmed approaches habit through the repeated use of objects and systems, whereas the psychology study focuses on behavioural routines and automatic responses that are developed through repetition over a sustained period. Altogether, these texts provide a useful contextual framing for understanding how users engage intuitively with audio technologies and interfaces in exhibitions (linking back to my project).

Ahmed argues that objects gain meaning through histories of repeated use. Stating; "they matter because they are regularly used by most as it has been for generations" (Ahmed, 2019), "Indeed, we know what they are because we know what they are for."(Ahmed, 2019). This statement depicts that familiarity is socially constructed by users overtime; objects begin to feel natural when being used because of their repeated use within daily life.

Supporting this, comes with the psychological perspective in Creature of Habit expanding this understanding by explaining how repeated actions become automatic behaviours. The article states that; "our daily lives involve high levels of repetition of activities within similar contexts" and that "habits are automatically triggered behaviours in which we engage without conscious awareness or deliberate control". Having these ideas together, familiarity is presented as not instinctive, but something that is developed through repeated interaction with, environments, routines, objects and systems.

This contextual framing connects directly to my project exploring audio augmented interaction and immersive sound within exhibitions. Visitors already understand how to use headphones, audio systems and sound interfaces observed by me in the exhibition and this is all because these technologies have become embedded within daily life.

That repeated exposure to personal audio devices has normalised their use psychologically and socially. This means interaction often feels intuitive even within unfamiliar environments such as museums.

Ahmed's argument helps explain how the object acquires meaning through a sustained history of use, while Creature of Habit explains how behavioural repetition creates automatic engagement with these systems.

Contextual Framing

However, the two texts also contrast in important ways. Creature of Habit focuses on routine behaviours that have a strong tie to environmental context. The article notes that “within a different environment, the same actions involved in making breakfast may suddenly run less smoothly, requiring conscious attention”. Suggesting habits can be disrupted when routines or surroundings are manipulated/changed.

On the other hand, Ahmed’s discussion of objects operates differently. While environments may change, the function and familiarity of certain objects can remain stable. I will refer to headphones again like for example; headphones being used similarly whether in a museum, on public transport or at home because Ahmed’s text focuses more on the use of objects showcasing that their meaning and purpose remain socially understood.

This distinction is important within UX and exhibition design because it highlights how both behaviour and object familiarity shape interaction. Audio augmented systems in exhibitions can feel intuitive not only because of interface design, but because users already carry psychological and social habits related to sound technologies to the point where they don’t have to read contextual information that gets provided in exhibitions.

Together, Ahmed’s theory of use and the psychological theory of habit provides a broader understanding of how familiarity, repetition and automatic behaviour can influence the engagement of visitors in interactive audio experiences.

Critical Analysis of Practice

Through my practice, I realised that Ahmed's theory was actively happening inside the exhibition space. Sara Ahmed's argument in What's the Use? On the Uses of Use suggests that objects acquire meaning not only from pure interface design but also through repeated and sustained use over time. This became highly relevant within my own research practice and observations at the exhibition. Visitors were able to engage intuitively with the jukebox and audio systems not necessarily because the interfaces were perfectly designed, but because prior histories of audio interaction already existed within users' everyday lives. This revealed that familiarity with sound technologies strongly shaped engagement within the exhibition space.

Visitors rarely needed instructions on how to use the jukebox artefact because the interaction already felt culturally familiar. While some visitors hesitated initially, many quickly understood that moving the dial would change the track. Once they recognised the function of this interaction, they were able to navigate the system with smoothly with ease. Reflecting Ahmed's statement that "we know what they are because we know what they are for" (Ahmed, 2019).

As a result, visitors were observed frequently skipping tracks, listening only partially to audio pieces, or selectively engaged with content that interested them. During observation, some visitors listened for only 15–43 seconds before moving onto another track.

Suggesting that familiarity is constructed gradually through repeated use, causing interactions with audio-augmented systems to feel intuitive even within unfamiliar environments such as exhibitions.

However, even though familiarity appeared to increase accessibility and make interaction easier, it also encouraged fragmented engagement. This is because curators may design immersive audio experiences to communicate structured narratives around identity and human experience however, visitor-controlled systems give users the agency to engage selectively, which potentially disrupts the intended narrative flow the curators aim to guide.

The jukebox tracks being a perfect example, may have been arranged in a meaningful sequence, whether chronological, thematic, or emotional. However, visitors often interacted with the system similarly to music-streaming platforms like Spotify; rapidly skipping between content based on curiosity or personal preference.

In addition to this, I believe that selective engagement may also deepen emotional resonance within visitors by giving them the freedom to prioritise content that feels personally or culturally relevant to them as people.

This reflects on my own experience within the exhibition, making me realise that if I were unable to control the tracks myself, I would likely lose interest waiting for multiple audio pieces to play through before reaching content I connected with initially. This demonstrates how visitor agency can influence dwell time and emotional engagement. In this sense, curiosity and personal relevance often become prioritised over curatorial sequencing.

Essay Section

Critical Analysis of Practice

Initially, my project focused broadly on interaction and immersion. However, through research, observation, and synthesis, I realised that familiarity and habitual audio behaviours were shaping visitor engagement more than expected.

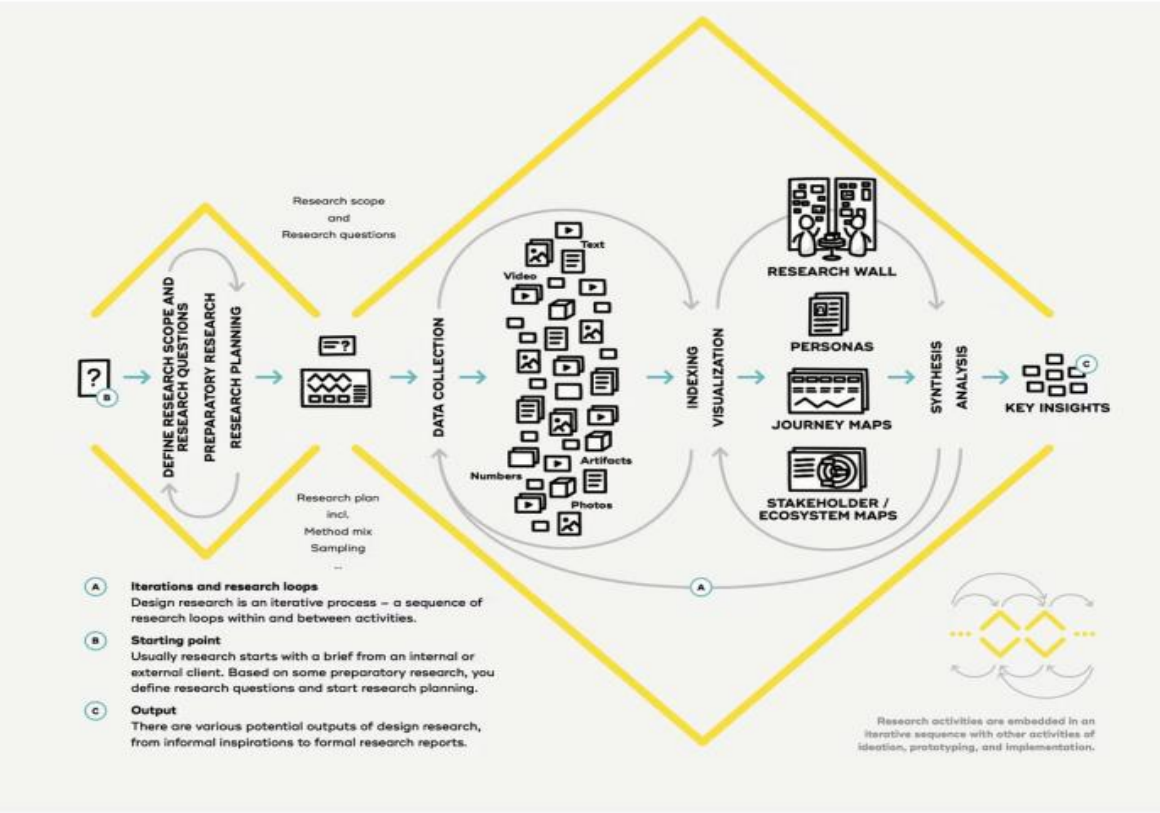
Ahmed's theory helped me reinterpret these interactions not simply as interface-based behaviours, but as socially learned patterns of use developed through repeated exposure to sound technologies in everyday life.

Ultimately, this research suggests that intuitive interaction is not created through design alone, but through the histories of repeated use that visitors bring into exhibition spaces.

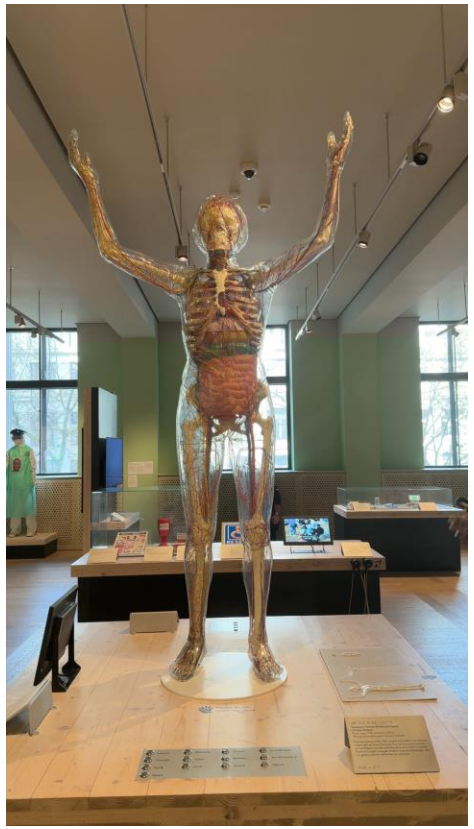
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Appendix



Screenshot about the Discover + Define section of the double diamond taken from a slide presentation created by my tutors which was taken from **Stickdorn, et al. (2018).**

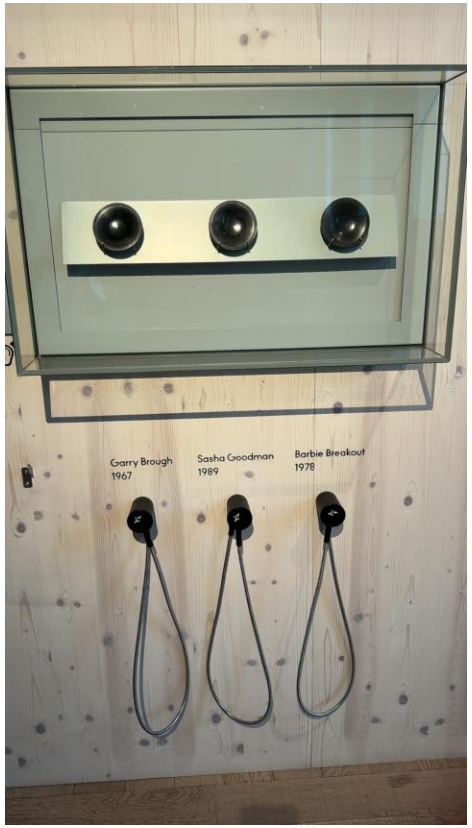


Transparent Woman



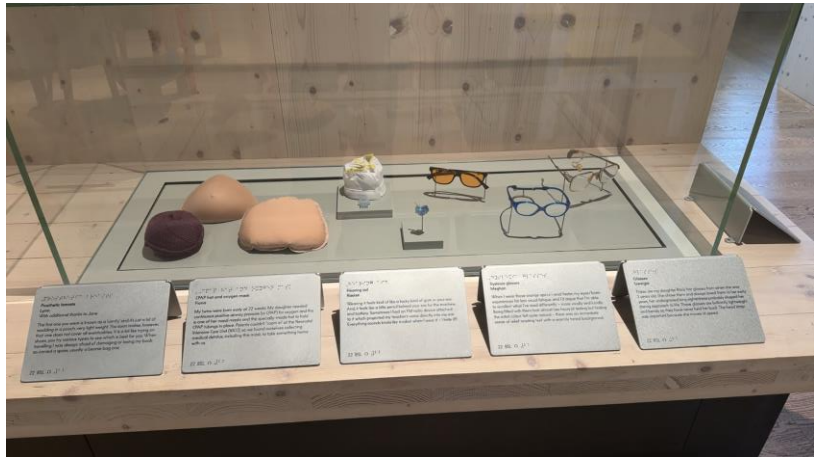
The Jukebox

Appendix

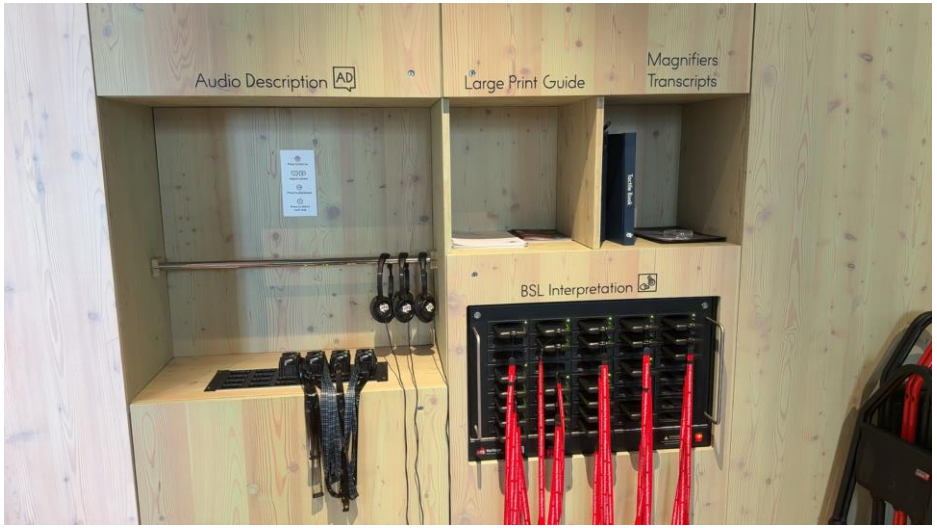


Blood Objects





Prosthetics

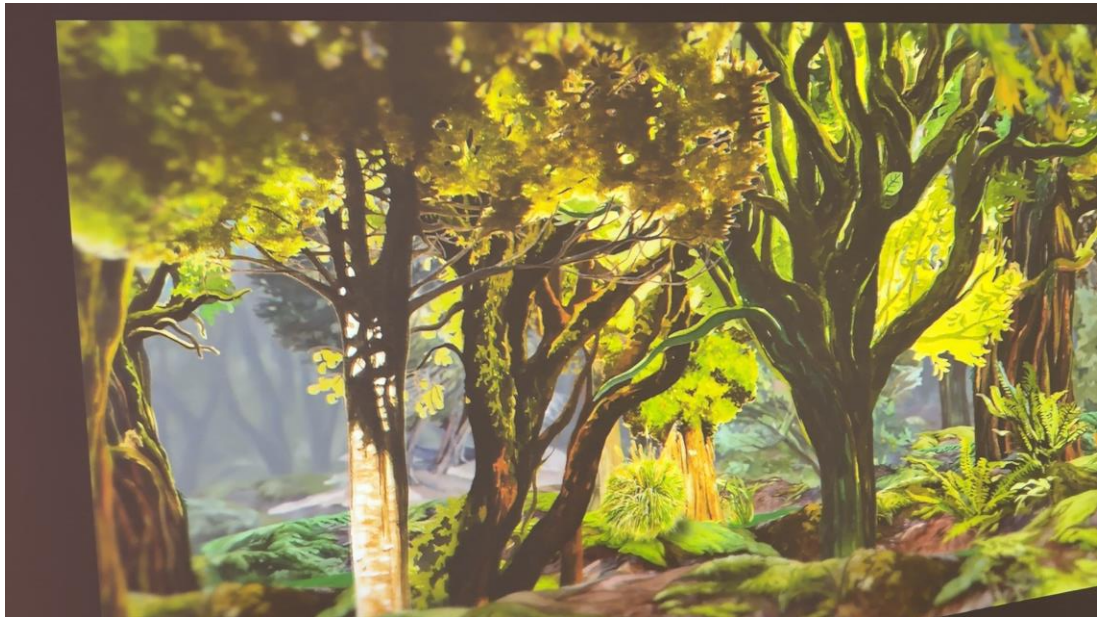






Accessibility + Signage

Appendix



The White



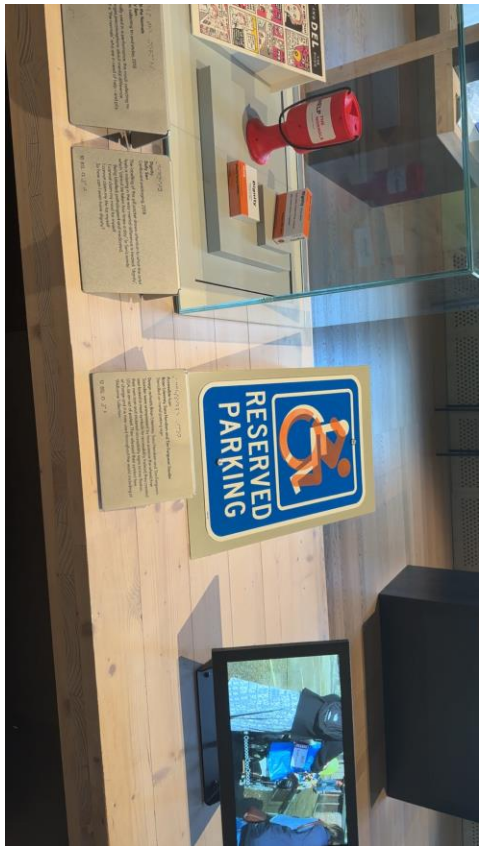
Disappearing message



Transparent woman buttons

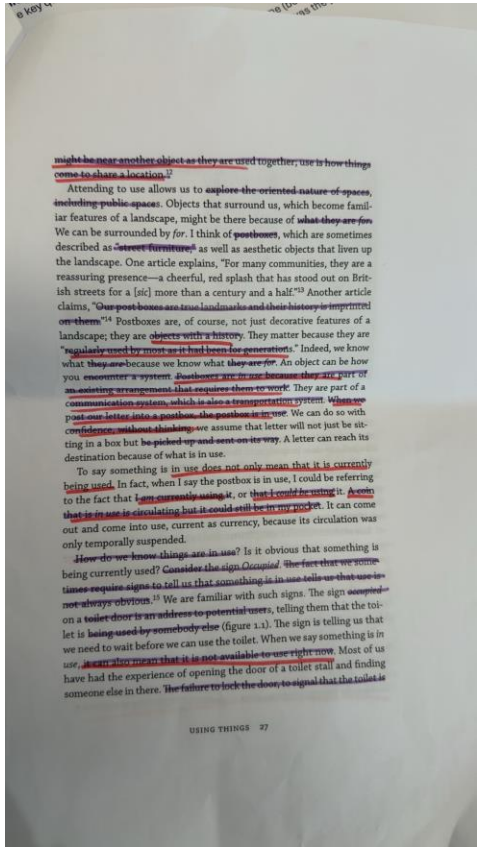
Appendix







More pictures of artefacts in the exhibition



might be near another object as they are used together; use is how things come to share a location.[12]

Attending to use allows us to explore the oriented nature of spaces, including public spaces. Objects that surround us, which become familiar features of a landscape, might be there because of what they are for. We can be surrounded by *for*. I think of postboxes, which are sometimes described as "street furniture,"[5] as well as aesthetic objects that liven up the landscape. One article explains, "For many communities, they are a reassuring presence—a cheerful, red splash that has stood out on British streets for a [sic] more than a century and a half."[13] Another article claims, "Our post boxes are true landmarks and their history is imprinted on them."[14] Postboxes are, of course, not just decorative features of a landscape; they are objects with a history. They matter because they are "regularly used by most as it had been for generations." Indeed, we know what they are because we know what they are *for*. An object can be how you encounter a system. Postboxes are in use because they are part of an existing arrangement that requires them to work. They are part of a communication system, which is also a transportation system. When we post our letter into a postbox, the postbox is in use. We can do so with confidence, without thinking, we assume that letter will not just be sitting in a box but be picked up and sent on its way. A letter can reach its destination because of what is in use.

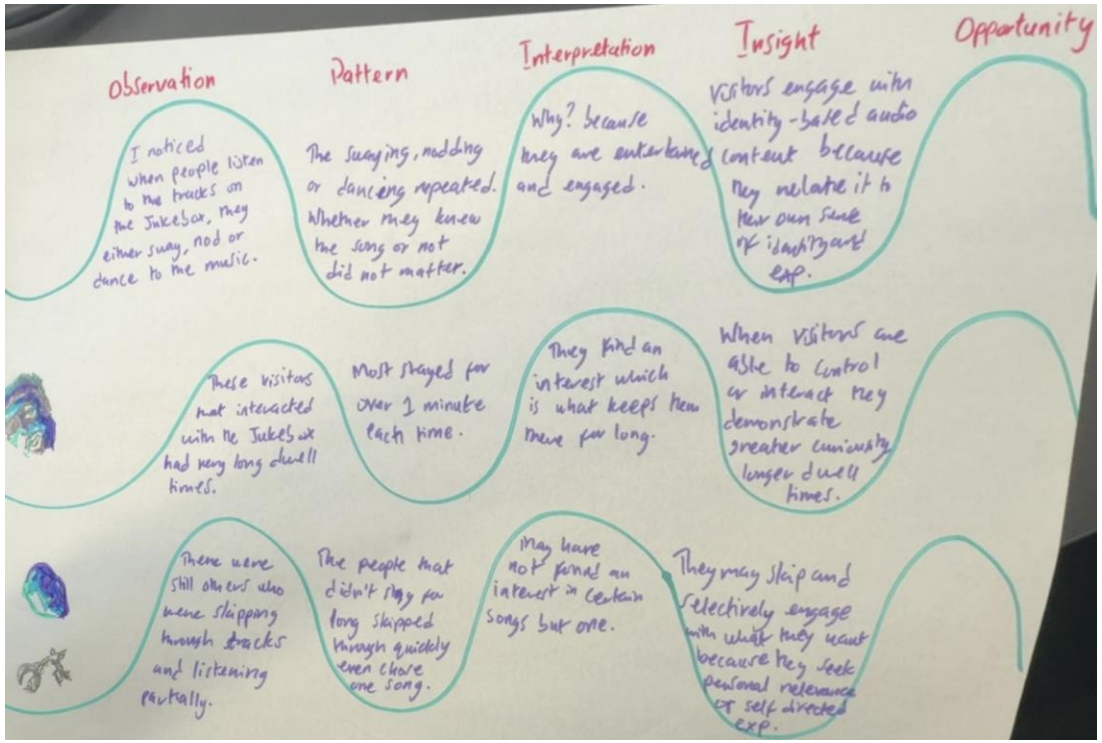
To say something is in use does not only mean that it is currently being used. In fact, when I say the postbox is in use, I could be referring to the fact that I am currently using it, or that I could be using it. A coin that is in use is circulating but it could still be in my pocket. It can come out and come into use, current as currency, because its circulation was only temporally suspended.

How do we know things are in use? Is it obvious that something is being currently used? Consider the sign *Occupied*. The fact that we sometimes require signs to tell us that something is in use tells us that use is not always obvious.[15] We are familiar with such signs. The sign *occupied* on a toilet door is an address to potential users, telling them that the toilet is being used by somebody else (figure 1.1). The sign is telling us that we need to wait before we can use the toilet. When we say something is *in use*, it can also mean that it is not available to use right now. Most of us have had the experience of opening the door of a toilet stall and finding someone else in there. The failure to lock the door, to signal that the toilet is

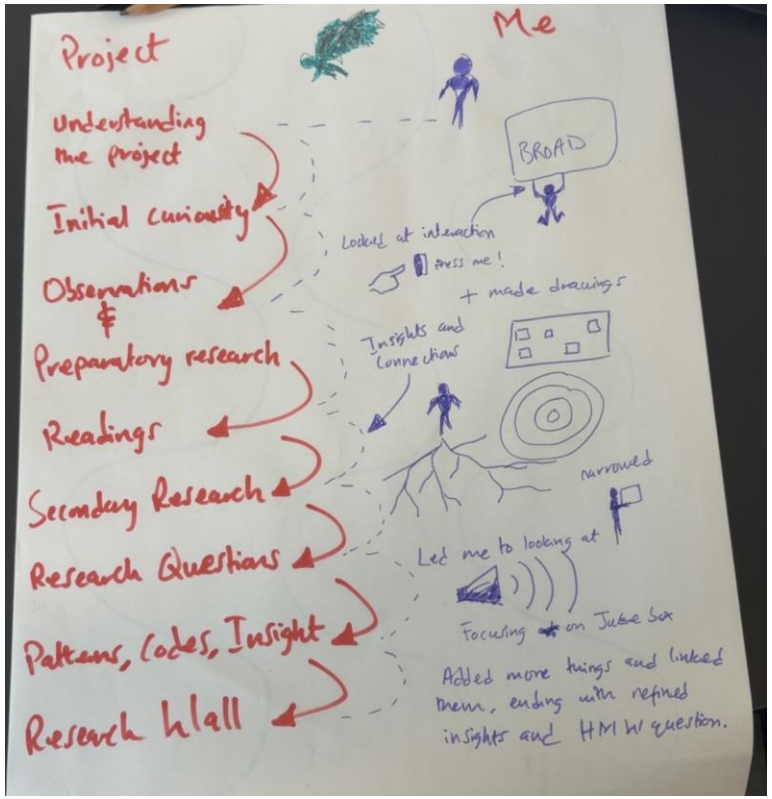
USING THINGS 27

Highlighting Sara Ahmed, What's the Use

Appendix



Opportunity framing + How might we draft



Tracking my progress, evaluating and reflection